

33. $a_1 = 1, a_{n+1} = \sqrt{2 + a_n}$

34. $a_1 = 100, a_{n+1} = \frac{1}{2} \left(a_n + \frac{25}{a_n} \right)$

35. Antibiotic pharmacokinetics A doctor prescribes a 100-mg antibiotic tablet to be taken every eight hours. Just before each tablet is taken, 20% of the drug present in the preceding time step remains in the body.

- How much of the drug is in the body just after the second tablet is taken? After the third tablet?
- If Q_n is the quantity of the antibiotic in the body just after the n th tablet is taken, write a difference equation that expresses Q_{n+1} in terms of Q_n .
- Find a formula for Q_n as a function of n .
- What quantity of the antibiotic remains in the body in the long run?

36. Drug pharmacokinetics A patient is injected with a drug every 12 hours. Immediately before each injection the concentration of the drug has been reduced by 90% and the new dose increases the concentration by 1.5 mg/mL.

- What is the concentration after three doses?
- If C_n is the concentration after the n th dose, write a difference equation that expresses C_{n+1} in terms of C_n .
- Find a formula for C_n as a function of n .
- What is the limiting value of the concentration?

37. Drug pharmacokinetics A patient takes 150 mg of a drug at the same time every day. Just before each tablet is taken, 5% of the drug present in the preceding time step remains in the body.

- What quantity of the drug is in the body after the third tablet? After the n th tablet?
- What quantity of the drug remains in the body in the long run?

38. Insulin injection After injection of a dose D of insulin, the concentration of insulin in a patient's system decays exponentially and so it can be written as De^{-at} , where t represents time in hours and a is a positive constant.

- If a dose D is injected every T hours, write an expression for the sum of the residual concentrations just before the $(n + 1)$ st injection.
- Determine the limiting pre-injection concentration.
- If the concentration of insulin must always remain at or above a critical value C , determine a minimal dosage D in terms of C , a , and T .

39. Let $x = 0.99999 \dots$

- Do you think that $x < 1$ or $x = 1$?
- Sum a geometric series to find the value of x .
- How many decimal representations does the number 1 have?
- Which numbers have more than one decimal representation?

40. A sequence is defined recursively by $a_n = (5 - n)a_{n-1}$, $a_1 = 1$. Find the sum of all the terms of the sequence.

41–46 Express the number as a ratio of integers.

41. $0.\overline{8} = 0.8888 \dots$

42. $0.\overline{46} = 0.46464646 \dots$

43. $2.\overline{516} = 2.516516516 \dots$

44. $10.\overline{135} = 10.135353535 \dots$

45. $1.53\overline{42}$

46. $7.\overline{12345}$

 **47–52 Logistic equation** Plot enough terms of the logistic difference equation $x_{t+1} = cx_t(1 - x_t)$ to see how the terms behave. Does the sequence appear to be convergent? If so, estimate the limit and then, assuming the limit exists, calculate its exact value. If not, describe the behavior of the terms.

47. $x_0 = 0.1, c = 2$

48. $x_0 = 0.8, c = 2.6$

49. $x_0 = 0.2, c = 3.2$

50. $x_0 = 0.4, c = 3.5$

51. $x_0 = 0.1, c = 3.8$

52. $x_0 = 0.6, c = 3.9$

 **53. Logistic equation: Dependence on initial values** Compare plots of the first 20 terms of the logistic equation $x_{t+1} = \frac{1}{4}x_t(1 - x_t)$ for the initial values $x_0 = 0.2$ and $x_0 = 0.2001$. When the initial value changes slightly, how does the solution change?

 **54. Logistic equation: Dependence on initial values** Repeat Exercise 53 for the equation $x_{t+1} = 4x_t(1 - x_t)$ and compare with the results of Exercise 53. [This behavior is another part of what it means to be *chaotic*.]

 **55–58 The Ricker equation** $x_{t+1} = cx_te^{-x_t}$ was introduced in Exercise 1.6.32. Plot enough terms of the Ricker equation to see how the terms behave. Does the sequence appear to be convergent? If so, estimate the limit and then, assuming the limit exists, calculate its exact value. If not, describe the behavior of the terms.

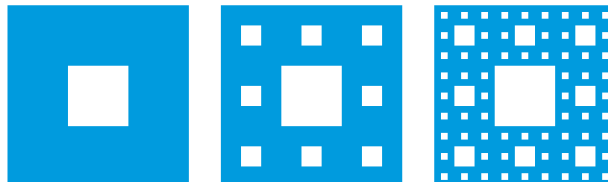
55. $x_0 = 0.2, c = 2$

56. $x_0 = 0.4, c = 3$

57. $x_0 = 0.8, c = 10$

58. $x_0 = 0.9, c = 20$

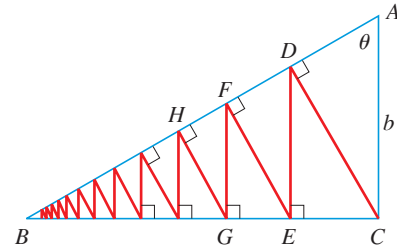
59. The Sierpinski carpet is constructed by removing the center one-ninth of a square of side 1, then removing the centers of the eight smaller remaining squares, and so on. (The figure shows the first three steps of the construction.) Show that the sum of the areas of the removed squares is 1. This implies that the Sierpinski carpet has area 0.



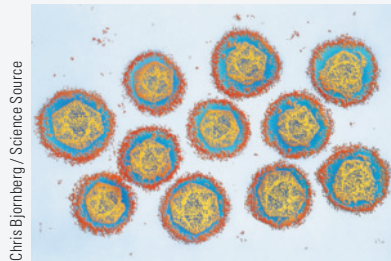
60. A right triangle ABC is given with $\angle A = \theta$ and $|AC| = b$. CD is drawn perpendicular to AB , DE is drawn perpendicular to BC , $EF \perp AB$, and this process is continued indefinitely, as shown in the figure. Find the total length of all the perpendiculars

$$|CD| + |DE| + |EF| + |FG| + \dots$$

in terms of b and θ .



PROJECT Modeling the Dynamics of Viral Infections¹



Chris Bjornberg / Science Source

A patient is infected with a virus that triples its numbers every hour. The immune system eventually kicks in and reduces the replication rate by a factor of $\frac{1}{2}$ in addition to killing 500,000 virus particles per hour, but this doesn't happen until the viral load reaches two million copies.

To combat the infection, the infected person receives hourly doses of an antiviral drug. This drug further reduces the replication rate, to a value of 1.25, and the immune system and the drug together can kill 25,000,000 copies of the virus per hour.

Let's model the phases of the infection using a discrete-time difference equation.

1. What is the recursion for the number of viral particles in the absence of treatment and before the immune response starts? What is the equation for the number of viral particles as a function of time?
2. How long does it take for the immune system to be activated after the infection starts? Derive an equation for this length of time for an arbitrary initial number of viral particles.
3. What is the recursion for the number of viral particles after the immune response has begun, but before the drug is used?
4. What is the condition for the viral population size to decrease over time solely because of the immune system? Is this possible?
5. What is the recursion for the number of viral particles in the presence of both the immune response and the drug?
6. What is the condition for the viral population size to decrease over time when the drug and immune system are both acting? Is this possible?
7. If an individual is infected by one virus, how much time do you have to start drug treatment in order to control the infection?
8. Often the outcome of an infection depends on the number of viral particles causing the infection. Derive an expression for the amount of time it takes from the initial infection for an individual to reach the critical viral load—beyond which control of the infection is impossible—as a function of the initial number of particles n_0 and an arbitrary initial replication rate R .

1. Adapted from F. Giordano et al., *A First Course in Mathematical Modeling* (Belmont, CA: Cengage Learning, 2014).

9. Suppose 24 hours have passed since the start of an infection with a single viral particle. How many viral particles are in the body? Can the use of the drug now control the infection? If so, how long will it take until the individual is free of the virus?

2.2 Limits of Functions at Infinity

In the preceding section we looked at the long-term behavior of a sequence $\{a_n\}$ by analyzing what happens to the terms a_n when n becomes large. Here we do something similar: We ask what happens to function values $f(x)$ when x becomes large. The only difference is that x is no longer restricted to be an integer.

The Monod Growth Function

In the 1930s and 1940s the French biologist Jacques Monod carried out experiments on *E. coli* bacteria feeding on a single nutrient, such as glucose. If N denotes the concentration of the nutrient, he modeled the per capita reproduction rate R of the bacteria as a function of N :

(1)
$$R(N) = \frac{SN}{c + N}$$

where c is a positive constant and S is the saturation level of the nutrient. The function $R(N)$ given by Equation 1 is called the **Monod growth function**. Monod later recognized that functions of this form had been used in biochemistry to model enzyme reactions, in which case the function is called the Michaelis-Menten function.

EXAMPLE 1 | **BB** The Monod growth function with $S = 2$ and $c = 5$ is

$$R(N) = \frac{2N}{5 + N}$$

Evaluate it for $N = 5, 10, 50, 100, 500, 1000, 5000, 10,000$. Then graph the function and comment on the shape of the graph.

SOLUTION The values are shown in the table and graphs of the function are shown on the intervals $[0, 10]$, $[0, 100]$, and $[0, 1000]$ in Figure 1.

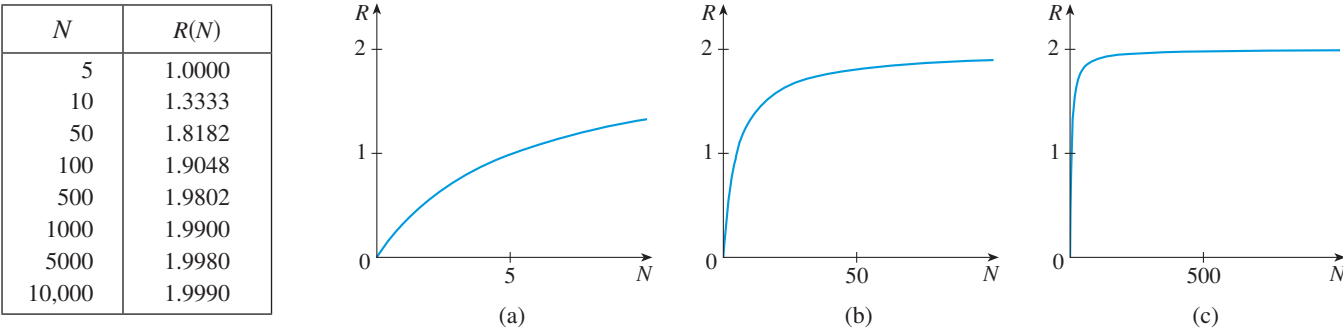


FIGURE 1

We see that $R(N)$ is an increasing function whose values are always less than 2 (the saturation level) but which approach 2 as N grows larger. Biologically, this means that the rate of reproduction of each bacterium increases with nutrient concentration, getting closer to 2 but never exceeding this value. ■

In Example 1 we saw that as N gets larger and larger the values of $R(N)$ become closer and closer to 2. In fact it seems that we can make the values of $R(N)$ as close as we like to 2 by making N sufficiently large. This situation is expressed symbolically by writing

$$\lim_{N \rightarrow \infty} R(N) = 2$$

■ Definition of a Limit at Infinity

In general, we use the notation

$$\lim_{x \rightarrow \infty} f(x) = L$$

to indicate that the values of $f(x)$ become closer and closer to L as x becomes larger and larger.

(2) Definition Let f be a function defined on some interval (a, ∞) . Then

$$\lim_{x \rightarrow \infty} f(x) = L$$

means that the values of $f(x)$ can be made arbitrarily close to L by taking x sufficiently large.

Another notation for $\lim_{x \rightarrow \infty} f(x) = L$ is

$$f(x) \rightarrow L \quad \text{as} \quad x \rightarrow \infty$$

The symbol ∞ does not represent a number. Nonetheless, the expression $\lim_{x \rightarrow \infty} f(x) = L$ is often read as

“the limit of $f(x)$, as x approaches infinity, is L ”

or “the limit of $f(x)$, as x becomes infinite, is L ”

or “the limit of $f(x)$, as x increases without bound, is L ”

The meaning of such phrases is given by Definition 2. A more precise definition is given in Appendix D.

Geometric illustrations of Definition 2 are shown in Figure 2. Notice that there are many ways for the graph of f to approach the line $y = L$ (which is called a *horizontal asymptote*) as we look to the far right of each graph.

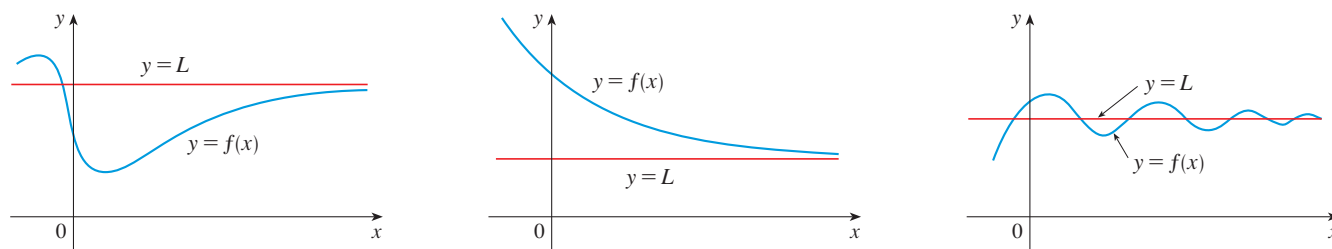
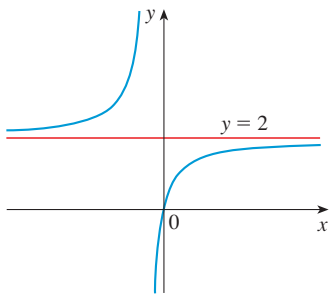


FIGURE 2 Examples illustrating $\lim_{x \rightarrow \infty} f(x) = L$

**FIGURE 3**

$$y = \frac{2x}{5 + x}$$

What happens if we let x decrease through negative values indefinitely? The Monod growth function has no biological meaning if N is negative, but let's consider the corresponding abstract mathematical function

$$f(x) = \frac{2x}{5 + x}$$

for numerically large negative values of x . The graph of f in Figure 3 shows that the values of $f(x)$ also approach 2 as x decreases through negative values without bound. This is expressed by writing

$$\lim_{x \rightarrow -\infty} \frac{2x}{5 + x} = 2$$

The general definition is as follows.

(3) Definition Let f be a function defined on some interval $(-\infty, a)$. Then

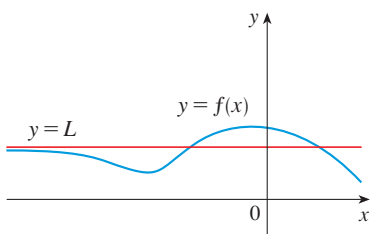
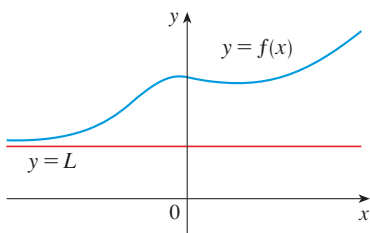
$$\lim_{x \rightarrow -\infty} f(x) = L$$

means that the values of $f(x)$ can be made arbitrarily close to L by taking x sufficiently large negative.

Again, the symbol $-\infty$ does not represent a number, but the expression $\lim_{x \rightarrow -\infty} f(x) = L$ is often read as

“the limit of $f(x)$, as x approaches negative infinity, is L ”

Definition 3 is illustrated in Figure 4. Notice that the graph approaches the line $y = L$ as we look to the far left of each graph.

**FIGURE 4**

Examples illustrating $\lim_{x \rightarrow -\infty} f(x) = L$

(4) Definition The line $y = L$ is called a **horizontal asymptote** of the curve $y = f(x)$ if either

$$\lim_{x \rightarrow \infty} f(x) = L \quad \text{or} \quad \lim_{x \rightarrow -\infty} f(x) = L$$

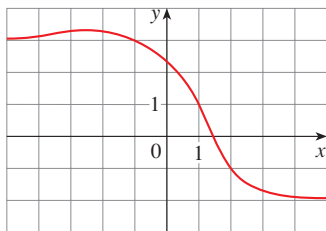
For instance, the curves in Figures 2 and 4 have the line $y = L$ as a horizontal asymptote.

EXAMPLE 2 | What are the horizontal asymptotes of the curve shown in Figure 5?

SOLUTION From the portion of the graph shown in Figure 5, it appears that $f(x)$ approaches -2 when x gets large and approaches 3 when x becomes large negative. So

$$\lim_{x \rightarrow \infty} f(x) = -2 \quad \text{and} \quad \lim_{x \rightarrow -\infty} f(x) = 3$$

This means that both $y = -2$ and $y = 3$ are horizontal asymptotes of the curve $y = f(x)$.

**FIGURE 5**

EXAMPLE 3 | Find $\lim_{x \rightarrow \infty} \frac{1}{x}$ and $\lim_{x \rightarrow -\infty} \frac{1}{x}$.

SOLUTION Observe that when x is large, $1/x$ is small. For instance,

$$\frac{1}{100} = 0.01 \quad \frac{1}{10,000} = 0.0001 \quad \frac{1}{1,000,000} = 0.000001$$

In fact, by taking x large enough, we can make $1/x$ as close to 0 as we please. Therefore, according to Definition 2, we have

$$\lim_{x \rightarrow \infty} \frac{1}{x} = 0$$

Similar reasoning shows that when x is large negative, $1/x$ is small negative, so we also have

$$\lim_{x \rightarrow -\infty} \frac{1}{x} = 0$$

It follows that the line $y = 0$ (the x -axis) is a horizontal asymptote of the curve $y = 1/x$. (See Figure 6.)

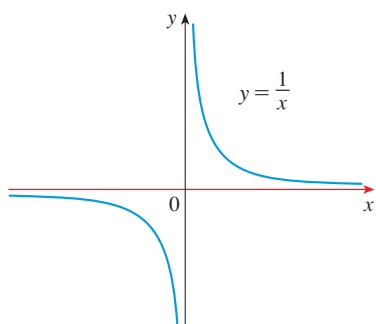


FIGURE 6

$$\lim_{x \rightarrow \infty} \frac{1}{x} = 0, \quad \lim_{x \rightarrow -\infty} \frac{1}{x} = 0$$

The limit laws for sequences that we used in Section 2.1 have their counterparts for functions. For instance, the Sum Law for functions states that

$$\lim_{x \rightarrow \infty} [f(x) + g(x)] = \lim_{x \rightarrow \infty} f(x) + \lim_{x \rightarrow \infty} g(x)$$

if these limits exist. And the following limit, which corresponds to Equation 2.1.2, is also very useful.

$$\lim_{x \rightarrow \infty} \frac{1}{x^p} = 0 \quad \text{for any number } p > 0$$

EXAMPLE 4 | **BB** The Monod growth function (continued) From Example 1 it appears that the limit of the Monod function is $\lim_{N \rightarrow \infty} R(N) = 2$. Verify this using the Limit Laws.

SOLUTION Dividing the numerator and denominator by N , and using the Limit Laws, we get

$$\begin{aligned} \lim_{N \rightarrow \infty} R(N) &= \lim_{N \rightarrow \infty} \frac{2N}{5 + N} = \lim_{N \rightarrow \infty} \frac{2}{\frac{5}{N} + 1} \\ &= \frac{2}{5 \lim_{N \rightarrow \infty} (1/N) + \lim_{N \rightarrow \infty} 1} = \frac{2}{5 \cdot 0 + 1} = 2 \end{aligned}$$

We conclude from Example 4 that the line $R = 2$ (corresponding to the saturation level) is a horizontal asymptote of the Monod function $R(N) = 2N/(5 + N)$. For the general Monod function given by Equation 1, we can show similarly that

$$\lim_{N \rightarrow \infty} R(N) = \lim_{N \rightarrow \infty} \frac{SN}{c + N} = S$$

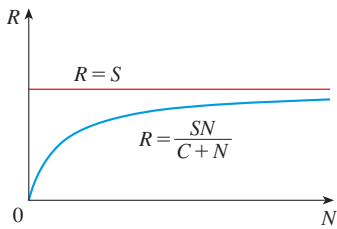


FIGURE 7

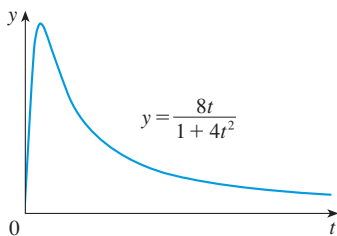


FIGURE 8

So the line $R = S$ is a horizontal asymptote (see Figure 7). Biologically, this means that no individual can have a reproductive rate greater than S , and this rate is approached asymptotically as the nutrient concentration becomes very large.

EXAMPLE 5 | Evaluate $\lim_{t \rightarrow \infty} \frac{8t}{1+4t^2}$.

SOLUTION As with any rational function, we first divide both the numerator and denominator by the highest power of t that occurs in the denominator:

$$\begin{aligned} \lim_{t \rightarrow \infty} \frac{8t}{1+4t^2} &= \lim_{t \rightarrow \infty} \frac{\frac{8t}{t^2}}{\frac{1}{t^2} + 4} = \lim_{t \rightarrow \infty} \frac{\frac{8}{t}}{\frac{1}{t^2} + 4} \\ &= \frac{8 \lim_{t \rightarrow \infty} \frac{1}{t}}{\lim_{t \rightarrow \infty} \frac{1}{t^2} + \lim_{t \rightarrow \infty} 4} = \frac{8 \cdot 0}{0 + 4} = 0 \end{aligned}$$

So the t -axis is a horizontal asymptote. Figure 8 shows the graph of the function $f(t) = 8t/(1+4t^2)$ for $t \geq 0$. If a biologist were looking for a function to model a quantity that starts at 0, quickly reaches a peak, and then gradually decays toward 0, then a function similar to f might be a suitable candidate. ■

EXAMPLE 6 | Compute $\lim_{x \rightarrow \infty} (\sqrt{x^2+1} - x)$.

SOLUTION Because both $\sqrt{x^2+1}$ and x are large when x is large, it's difficult to see what happens to their difference, so we use algebra to rewrite the function.

We first multiply numerator and denominator by the conjugate radical:

$$\begin{aligned} \lim_{x \rightarrow \infty} (\sqrt{x^2+1} - x) &= \lim_{x \rightarrow \infty} (\sqrt{x^2+1} - x) \frac{\sqrt{x^2+1} + x}{\sqrt{x^2+1} + x} \\ &= \lim_{x \rightarrow \infty} \frac{(x^2+1) - x^2}{\sqrt{x^2+1} + x} = \lim_{x \rightarrow \infty} \frac{1}{\sqrt{x^2+1} + x} \end{aligned}$$

Notice that the denominator of this last expression $(\sqrt{x^2+1} + x)$ becomes large as $x \rightarrow \infty$ (it's bigger than x). So

$$\lim_{x \rightarrow \infty} (\sqrt{x^2+1} - x) = \lim_{x \rightarrow \infty} \frac{1}{\sqrt{x^2+1} + x} = 0$$

Figure 9 illustrates this result. ■

We can think of the given function as having a denominator of 1.

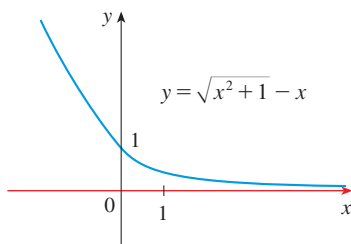


FIGURE 9

EXAMPLE 7 | Evaluate $\lim_{x \rightarrow \infty} \sin x$.

SOLUTION As x increases, the values of $\sin x$ oscillate between 1 and -1 infinitely often and so they don't approach any definite number. Thus $\lim_{x \rightarrow \infty} \sin x$ does not exist. ■

■ Limits Involving Exponential Functions

Let's recall the shapes of the graphs of the exponential functions from Figures 1.4.4 and 1.4.5. The graphs take the general shape shown in Figure 10.

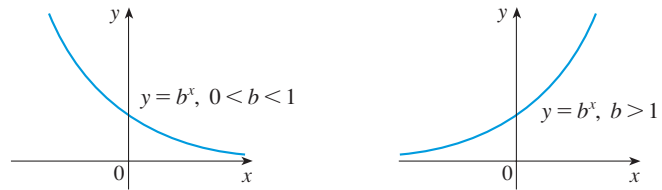


FIGURE 10

In both cases ($0 < b < 1$ and $b > 1$) the line $y = 0$ is a horizontal asymptote. Specifically:

$$\text{If } 0 < b < 1, \quad \text{then} \quad \lim_{x \rightarrow \infty} b^x = 0.$$

$$\text{If } b > 1, \quad \text{then} \quad \lim_{x \rightarrow -\infty} b^x = 0.$$

For instance, in Section 1.4 we considered the model

$$(5) \quad V(t) = 96.39785 \cdot (0.818656)^t$$

for the viral load in a patient with HIV. Here the base of the exponential function is $b = 0.818656$, which is less than 1, and so

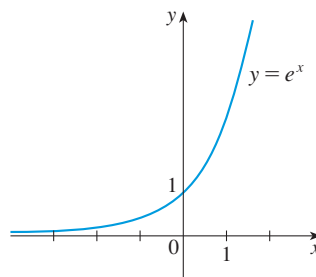
$$\lim_{t \rightarrow \infty} V(t) = 0$$

which is confirmed in Figure 1.4.12.

For the most important special case, $b = e$, we have the graph in Figure 11, illustrating the fact that

$$\lim_{x \rightarrow -\infty} e^x = 0$$

You can see that the values of e^x approach 0 very rapidly as $x \rightarrow -\infty$.



x	e^x
0	1.00000
-1	0.36788
-2	0.13534
-3	0.04979
-5	0.00674
-8	0.00034
-10	0.00005

FIGURE 11

Recall from Section 1.3 that we get the graph of $y = f(-x)$ by reflecting the graph of $y = f(x)$ about the y -axis. So the graph of the function $y = e^{-x}$ is obtained by reflecting the graph of $y = e^x$ (Figure 11) about the y -axis. This function occurs so often in biology that it's wise to be familiar with the shape of its graph and its behavior for large x :

$$(6) \quad \lim_{x \rightarrow \infty} e^{-x} = 0$$

Another way to see the limit of the viral load function $V(t)$ in Equation 5 is to write $V(t)$ in terms of a power of e . In general, $b^x = (e^{\ln b})^x = e^{x \ln b}$ and so, with $b = 0.818656$ and using Equation 6, we have

$$\lim_{t \rightarrow \infty} V(t) = \lim_{t \rightarrow \infty} (96.39785 \cdot e^{-0.2001t}) = 0$$

Gause

G. F. Gause (1910–1986) was a Russian biologist whose success was a result of his training and expertise in both mathematics and experimental biology. He proposed the *competitive exclusion principle*: Two species competing for the same resources cannot coexist if one has even a slight advantage over the other.

EXAMPLE 8 | **BB** **Gause's logistic model** In the 1930s the biologist G. F. Gause conducted an experiment with the protozoan *Paramecium*. He modeled his data with the logistic function

$$P(t) = \frac{64}{1 + 31e^{-0.7944t}}$$

for the protozoan population after t days. Find the initial population and the limiting population.

SOLUTION The initial population was

$$P(0) = \frac{64}{1 + 31 \cdot e^0} = 2$$

The limiting population was

$$\begin{aligned} \lim_{t \rightarrow \infty} P(t) &= \lim_{t \rightarrow \infty} \frac{64}{1 + 31e^{-0.7944t}} = \frac{64}{1 + 31 \lim_{t \rightarrow \infty} e^{-0.7944t}} \\ &= \frac{64}{1 + 31 \cdot 0} = 64 \end{aligned}$$

Notice that $-0.7944t \rightarrow -\infty$ as $t \rightarrow \infty$.

Here we have used the Limit Laws as well as Equation 6. ■

■ Infinite Limits at Infinity

The notation

$$\lim_{x \rightarrow \infty} f(x) = \infty$$

is used to indicate that the values of $f(x)$ become large as x becomes large. Similar meanings are attached to the following symbols:

$$\lim_{x \rightarrow -\infty} f(x) = \infty \quad \lim_{x \rightarrow \infty} f(x) = -\infty \quad \lim_{x \rightarrow -\infty} f(x) = -\infty$$

EXAMPLE 9 | Find $\lim_{x \rightarrow \infty} x^3$ and $\lim_{x \rightarrow -\infty} x^3$.

SOLUTION When x becomes large, x^3 also becomes large. For instance,

$$10^3 = 1000 \quad 100^3 = 1,000,000 \quad 1000^3 = 1,000,000,000$$

In fact, we can make x^3 as big as we like by taking x large enough. Therefore we can write

$$\lim_{x \rightarrow \infty} x^3 = \infty$$

Similarly, when x is large negative, so is x^3 . Thus

$$\lim_{x \rightarrow -\infty} x^3 = -\infty$$

These limit statements can also be seen from the graph of $y = x^3$ in Figure 12. ■

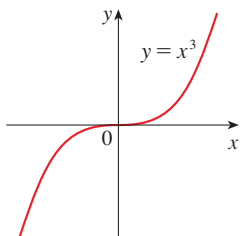


FIGURE 12

$$\lim_{x \rightarrow \infty} x^3 = \infty, \quad \lim_{x \rightarrow -\infty} x^3 = -\infty$$

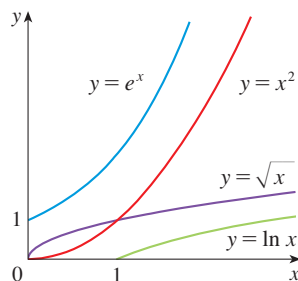


FIGURE 13

Looking at Figure 11 on page 107 we see that

$$\lim_{x \rightarrow \infty} e^x = \infty$$

It is also true that

$$\lim_{x \rightarrow \infty} x^2 = \infty$$

$$\lim_{x \rightarrow \infty} \sqrt{x} = \infty$$

$$\lim_{x \rightarrow \infty} \ln x = \infty$$

But as Figure 13 demonstrates, these four functions become large at different rates. In Chapter 4 we will see how to rank functions according to how quickly they grow.

EXAMPLE 10 | Find $\lim_{x \rightarrow \infty} (x^2 - x)$.

SOLUTION It would be **wrong** to write



$$\lim_{x \rightarrow \infty} (x^2 - x) = \lim_{x \rightarrow \infty} x^2 - \lim_{x \rightarrow \infty} x = \infty - \infty$$

The Limit Laws can't be applied to infinite limits because ∞ is not a number ($\infty - \infty$ can't be defined). However, we *can* write

$$\lim_{x \rightarrow \infty} (x^2 - x) = \lim_{x \rightarrow \infty} x(x - 1) = \infty$$

because both x and $x - 1$ become arbitrarily large and so their product does too. ■

EXAMPLE 11 | Find $\lim_{x \rightarrow \infty} \frac{x^2 + x}{3 - x}$.

SOLUTION As in Example 5, we divide the numerator and denominator by the highest power of x in the denominator, which is just x :

$$\lim_{x \rightarrow \infty} \frac{x^2 + x}{3 - x} = \lim_{x \rightarrow \infty} \frac{x + 1}{\frac{3}{x} - 1} = -\infty$$

because $x + 1 \rightarrow \infty$ and $3/x - 1 \rightarrow -1$ as $x \rightarrow \infty$. ■

EXERCISES 2.2

1. Explain in your own words the meaning of each of the following.

(a) $\lim_{x \rightarrow \infty} f(x) = 5$

(b) $\lim_{x \rightarrow -\infty} f(x) = 3$

2. (a) Can the graph of $y = f(x)$ intersect a horizontal asymptote? If so, how many times? Illustrate by sketching graphs.
 (b) How many horizontal asymptotes can the graph of $y = f(x)$ have? Sketch graphs to illustrate the possibilities.



3. Guess the value of the limit

$$\lim_{x \rightarrow \infty} \frac{x^2}{2^x}$$

by evaluating the function $f(x) = x^2/2^x$ for $x = 0, 1, 2, 3,$

4, 5, 6, 7, 8, 9, 10, 20, 50, and 100. Then use a graph of f to support your guess.



4. (a) Use a graph of

$$f(x) = \left(1 - \frac{2}{x}\right)^x$$

to estimate the value of $\lim_{x \rightarrow \infty} f(x)$ correct to two decimal places.

- (b) Use a table of values of $f(x)$ to estimate the limit to four decimal places.

5–28 Find the limit.

5. $\lim_{x \rightarrow \infty} \frac{1}{2x + 3}$

6. $\lim_{x \rightarrow \infty} \frac{3x + 5}{x - 4}$

7. $\lim_{x \rightarrow \infty} \frac{3x - 2}{2x + 1}$

8. $\lim_{x \rightarrow \infty} \frac{1 - x^2}{x^3 - x + 1}$

9. $\lim_{x \rightarrow \infty} \frac{1 - x - x^2}{2x^2 - 7}$

10. $\lim_{x \rightarrow -\infty} \frac{4x^3 + 6x^2 - 2}{2x^3 - 4x + 5}$

11. $\lim_{t \rightarrow -\infty} 0.6^t$

12. $\lim_{r \rightarrow \infty} \frac{5}{10^r}$

13. $\lim_{t \rightarrow \infty} \frac{\sqrt{t} + t^2}{2t - t^2}$

14. $\lim_{t \rightarrow \infty} \frac{t - t\sqrt{t}}{2t^{3/2} + 3t - 5}$

15. $\lim_{x \rightarrow \infty} \frac{(2x^2 + 1)^2}{(x - 1)^2(x^2 + x)}$

16. $\lim_{x \rightarrow \infty} \frac{x^2}{\sqrt{x^4 + 1}}$

17. $\lim_{x \rightarrow \infty} (\sqrt{9x^2 + x} - 3x)$

18. $\lim_{x \rightarrow \infty} (\sqrt{x^2 + ax} - \sqrt{x^2 + bx})$

19. $\lim_{x \rightarrow \infty} \frac{6}{3 + e^{-2x}}$

20. $\lim_{x \rightarrow \infty} \sqrt{x^2 + 1}$

21. $\lim_{x \rightarrow \infty} \frac{x^4 - 3x^2 + x}{x^3 - x + 2}$

22. $\lim_{x \rightarrow \infty} (e^{-x} + 2 \cos 3x)$

23. $\lim_{x \rightarrow -\infty} (x^4 + x^5)$

24. $\lim_{x \rightarrow -\infty} \frac{1 + x^6}{x^4 + 1}$

25. $\lim_{t \rightarrow \infty} e^{-1/t^2}$

26. $\lim_{x \rightarrow \infty} \frac{e^{3x} - e^{-3x}}{e^{3x} + e^{-3x}}$

27. $\lim_{x \rightarrow \infty} \frac{1 - e^x}{1 + 2e^x}$

28. $\lim_{x \rightarrow -\infty} [\ln(x^2) - \ln(x^2 + 1)]$

29. For the **Monod growth function** $R(N) = SN/(c + N)$, what is the significance of the constant c ? [Hint: What is $R(c)$?]

30. The **Michaelis-Menten equation** models the rate v of an enzymatic reaction as a function of the concentration $[S]$ of a substrate S . In the case of the enzyme chymotrypsin the equation is

$$v = \frac{0.14[S]}{0.015 + [S]}$$

- (a) What is the horizontal asymptote of the graph of v ? What is its significance?



- (b) Use a graphing calculator or computer to graph v as a function of $[S]$.

31. **Virulence and pathogen transmission** The number of new infections produced by an individual infected with a pathogen such as influenza depends on the mortality rate that the pathogen causes. This pathogen-induced mortality rate is referred to as the pathogen's *virulence*. [The photo

shows victims at Camp Funston of the influenza epidemic of 1918.] Extremely high levels of virulence result in very little transmission because the infected individual dies before infecting other individuals. Under certain assumptions, the number of new infections N is related to virulence v by the function

$$N(v) = \frac{8v}{1 + 2v + v^2}$$

where v is the mortality rate (that is, virulence) and $v \geq 0$. Evaluate $\lim_{v \rightarrow \infty} N(v)$ and interpret your result.



Science Source

32. The von Bertalanffy growth function

$$L(t) = L_{\infty} - (L_{\infty} - L_0)e^{-kt}$$

where k is a positive constant, models the length L of a fish as a function of t , the age of the fish. This model assumes that the fish has a well-defined length L_0 at birth ($t = 0$).

- (a) Calculate $\lim_{t \rightarrow \infty} L(t)$. How do you interpret the answer?
 (b) If $L_0 = 2$ cm and $L_{\infty} = 40$ cm, graph $L(t)$ for several values of k . What role does k play?



33. The Pacific halibut fishery has been modeled by the equation

$$B(t) = \frac{8 \times 10^7}{1 + 3e^{-0.71t}}$$

where $B(t)$ is the biomass (the total mass of the members of the population) in kilograms at time t . What is $\lim_{t \rightarrow \infty} B(t)$? What is the significance of this limit?

34. (a) A tank contains 5000 L of pure water. Brine that contains 30 g of salt per liter of water is pumped into the tank at a rate of 25 L/min. Show that the concentration of salt after t minutes (in grams per liter) is

$$C(t) = \frac{30t}{200 + t}$$

- (b) What happens to the concentration as $t \rightarrow \infty$?

35. Since $\lim_{x \rightarrow \infty} e^{-x} = 0$, we should be able to make e^{-x} as small as we like by choosing x large enough. How large do we have to take x so that $e^{-x} < 0.0001$?
36. Let $f(x) = x/(x + 1)$. What is $\lim_{x \rightarrow \infty} f(x)$? How large does x have to be so that $f(x) > 0.99$?
37. The velocity $v(t)$ of a falling raindrop at time t is modeled by the equation

$$v(t) = v^*(1 - e^{-gt/v^*})$$

where g is the acceleration due to gravity and v^* is the terminal velocity of the raindrop.

- (a) Find $\lim_{t \rightarrow \infty} v(t)$.
- (b) For a large raindrop in moderate rainfall, a typical terminal velocity is 7.5 m/s. How long does it take for the velocity of such a raindrop to reach 99% of its terminal velocity? (Take $g = 9.8 \text{ m/s}^2$.)

2.3 Limits of Functions at Finite Numbers

We've investigated what happens to function values $f(x)$ as x becomes large. Here we focus on what happens to $f(x)$ when x approaches a finite number. We begin by observing how such limits arise when we try to determine the speed of a falling ball. Later we'll see how similar considerations are involved in finding rates of change in biology.

Velocity Is a Limit

If you watch the speedometer of a car as you travel in city traffic, you see that the speed doesn't stay the same for very long; that is, the velocity of the car is not constant. We assume from watching the speedometer that the car has a definite velocity at each moment, but how is the "instantaneous" velocity defined? Let's investigate the example of a falling ball.



Steve Allen / Stockbyte / Getty Images

The CN Tower in Toronto was the tallest freestanding building in the world for 32 years.

EXAMPLE 1 | Suppose that a ball is dropped from the upper observation deck of the CN Tower in Toronto, 450 m above the ground. Find the velocity of the ball after 5 seconds.

SOLUTION Through experiments carried out four centuries ago, Galileo discovered that the distance fallen by any freely falling body is proportional to the square of the time it has been falling. (This model for free fall neglects air resistance.) If the distance fallen after t seconds is denoted by $s(t)$ and measured in meters, then Galileo's law is expressed by the equation

$$s(t) = 4.9t^2$$

The difficulty in finding the velocity after 5 s is that we are dealing with a single instant of time ($t = 5$), so no time interval is involved. However, we can approximate the desired quantity by computing the average velocity over the brief time interval of a tenth of a second from $t = 5$ to $t = 5.1$:

$$\begin{aligned} \text{average velocity} &= \frac{\text{change in position}}{\text{time elapsed}} \\ &= \frac{s(5.1) - s(5)}{0.1} \\ &= \frac{4.9(5.1)^2 - 4.9(5)^2}{0.1} = 49.49 \text{ m/s} \end{aligned}$$

The following table shows the results of similar calculations of the average velocity over successively smaller time periods.

Time interval	Average velocity (m/s)
$5 \leq t \leq 6$	53.9
$5 \leq t \leq 5.1$	49.49
$5 \leq t \leq 5.05$	49.245
$5 \leq t \leq 5.01$	49.049
$5 \leq t \leq 5.001$	49.0049

It appears that as we shorten the time period, the average velocity is becoming closer to 49 m/s. The **instantaneous velocity** when $t = 5$ is defined to be the limiting value of these average velocities over shorter and shorter time periods that start at $t = 5$. Thus the (instantaneous) velocity after 5 s appears to be

$$v = 49 \text{ m/s}$$

■ Limits: Numerical and Graphical Methods

We saw in Example 1 how limiting values arise in determining the velocity of an object. In Chapter 3 we will see how limits also occur in finding tangent lines to curves and rates of growth in biology, so we now turn our attention to limits in general and numerical and graphical methods for computing them.

Let’s investigate the behavior of the function f defined by $f(x) = x^2 - x + 2$ for values of x near 2. The following table gives values of $f(x)$ for values of x close to 2 but not equal to 2.

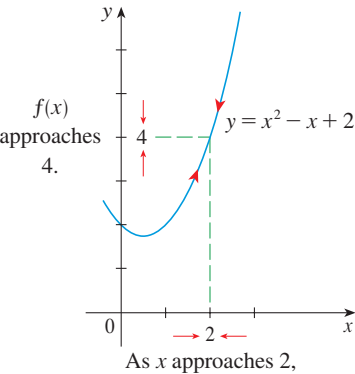


FIGURE 1

x	$f(x)$	x	$f(x)$
1.0	2.000000	3.0	8.000000
1.5	2.750000	2.5	5.750000
1.8	3.440000	2.2	4.640000
1.9	3.710000	2.1	4.310000
1.95	3.852500	2.05	4.152500
1.99	3.970100	2.01	4.030100
1.995	3.985025	2.005	4.015025
1.999	3.997001	2.001	4.003001

From the table and the graph of f (a parabola) shown in Figure 1 we see that when x is close to 2 (on either side of 2), $f(x)$ is close to 4. In fact, it appears that we can make the values of $f(x)$ as close as we like to 4 by taking x sufficiently close to 2. We express this by saying “the limit of the function $f(x) = x^2 - x + 2$ as x approaches 2 is equal to 4.” The notation for this is

$$\lim_{x \rightarrow 2} (x^2 - x + 2) = 4$$

In general, we use the following notation.

(1) Definition Suppose $f(x)$ is defined when x is near the number a . (This means that f is defined on some open interval that contains a , except possibly at a itself.) Then we write

$$\lim_{x \rightarrow a} f(x) = L$$

and say “the limit of $f(x)$, as x approaches a , equals L ”

if we can make the values of $f(x)$ arbitrarily close to L (as close to L as we like) by taking x to be sufficiently close to a (on either side of a) but not equal to a .

Roughly speaking, this says that the values of $f(x)$ approach L as x approaches a . In other words, the values of $f(x)$ tend to get closer and closer to the number L as x gets closer and closer to the number a (from either side of a) but $x \neq a$. (A more precise definition is given in Appendix D.)

An alternative notation for

$$\lim_{x \rightarrow a} f(x) = L$$

is $f(x) \rightarrow L$ as $x \rightarrow a$

which is usually read “ $f(x)$ approaches L as x approaches a .”

Notice the phrase “but $x \neq a$ ” in the definition of a limit. This means that in finding the limit of $f(x)$ as x approaches a , we never consider $x = a$. In fact, $f(x)$ need not even be defined when $x = a$. The only thing that matters is how f is defined *near* a .

Figure 2 shows the graphs of three functions. Note that in part (c), $f(a)$ is not defined and in part (b), $f(a) \neq L$. But in each case, regardless of what happens at a , it is true that $\lim_{x \rightarrow a} f(x) = L$.

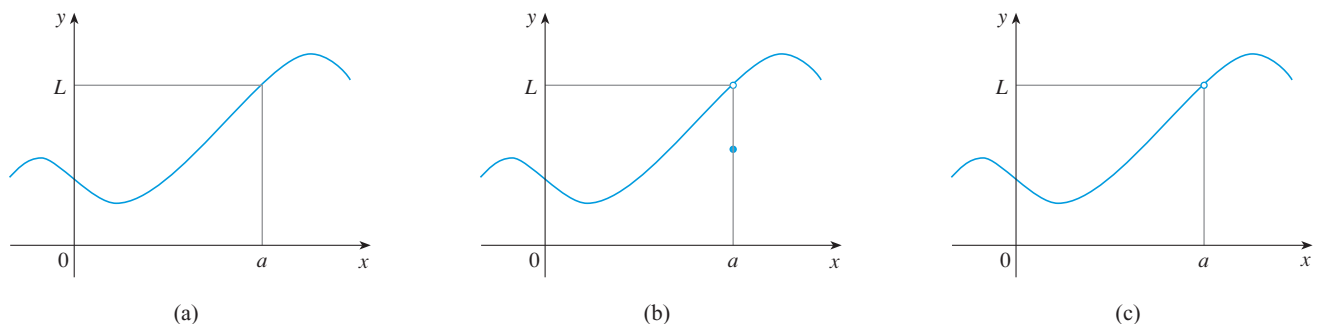


FIGURE 2
 $\lim_{x \rightarrow a} f(x) = L$ in all three cases.

EXAMPLE 2 | Guess the value of $\lim_{x \rightarrow 1} \frac{x-1}{x^2-1}$.

SOLUTION Notice that the function $f(x) = (x-1)/(x^2-1)$ is not defined when $x = 1$, but that doesn't matter because the definition of $\lim_{x \rightarrow a} f(x)$ says that we consider values of x that are close to a but not equal to a .

$x < 1$	$f(x)$
0.5	0.666667
0.9	0.526316
0.99	0.502513
0.999	0.500250
0.9999	0.500025

$x > 1$	$f(x)$
1.5	0.400000
1.1	0.476190
1.01	0.497512
1.001	0.499750
1.0001	0.499975

The tables at the left give values of $f(x)$ (correct to six decimal places) for values of x that approach 1 (but are not equal to 1). On the basis of the values in the tables, we make the guess that

$$\lim_{x \rightarrow 1} \frac{x - 1}{x^2 - 1} = 0.5$$

Example 2 is illustrated by the graph of f in Figure 3. Now let's change f slightly by giving it the value 2 when $x = 1$ and calling the resulting function g :

$$g(x) = \begin{cases} \frac{x - 1}{x^2 - 1} & \text{if } x \neq 1 \\ 2 & \text{if } x = 1 \end{cases}$$

This new function g still has the same limit as x approaches 1 (see Figure 4).

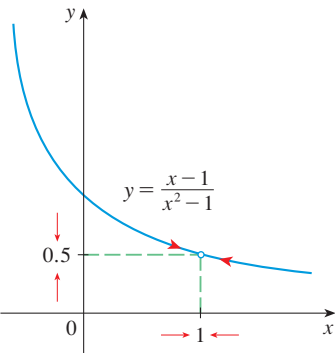


FIGURE 3

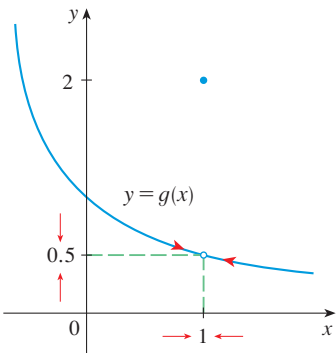


FIGURE 4

EXAMPLE 3 | Estimate the value of $\lim_{t \rightarrow 0} \frac{\sqrt{t^2 + 9} - 3}{t^2}$.

SOLUTION The table lists values of the function for several values of t near 0.

t	$\frac{\sqrt{t^2 + 9} - 3}{t^2}$
± 1.0	0.16228
± 0.5	0.16553
± 0.1	0.16662
± 0.05	0.16666
± 0.01	0.16667

As t approaches 0, the values of the function seem to approach 0.166666 . . . and so we guess that

$$\lim_{t \rightarrow 0} \frac{\sqrt{t^2 + 9} - 3}{t^2} = \frac{1}{6}$$

t	$\frac{\sqrt{t^2 + 9} - 3}{t^2}$
± 0.0005	0.16800
± 0.0001	0.20000
± 0.00005	0.00000
± 0.00001	0.00000

In Example 3 what would have happened if we had taken even smaller values of t ? The table in the margin shows the results from one calculator; you can see that something strange seems to be happening.

If you try these calculations on your own calculator you might get different values, but eventually you will get the value 0 if you make t sufficiently small. Does this mean that the answer is really 0 instead of $\frac{1}{6}$? No, the value of the limit is $\frac{1}{6}$, as we will show in the next section. The problem is that the **calculator gave false values** because $\sqrt{t^2 + 9}$ is very close to 3 when t is small. (In fact, when t is sufficiently small, a calculator's value for $\sqrt{t^2 + 9}$ is 3.000... to as many digits as the calculator is capable of carrying.)

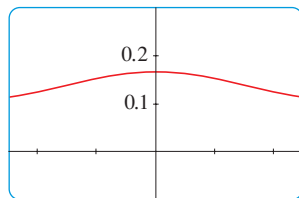
Something similar happens when we try to graph the function

$$f(t) = \frac{\sqrt{t^2 + 9} - 3}{t^2}$$

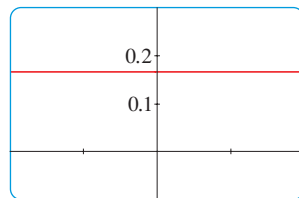
www.stewartcalculus.com

For a further explanation of why calculators sometimes give false values, click on *Lies My Calculator and Computer Told Me*. In particular, see the section called *The Perils of Subtraction*.

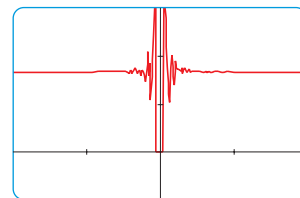
of Example 3 on a graphing calculator or computer. Parts (a) and (b) of Figure 5 show quite accurate graphs of f , and when we use the trace mode (if available) we can estimate easily that the limit is about $\frac{1}{6}$. But if we zoom in too much, as in parts (c) and (d), then we get inaccurate graphs, again because of problems with subtraction.



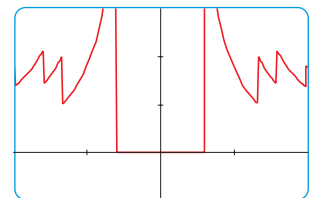
(a) $[-5, 5]$ by $[-0.1, 0.3]$



(b) $[-0.1, 0.1]$ by $[-0.1, 0.3]$



(c) $[-10^{-6}, 10^{-6}]$ by $[-0.1, 0.3]$



(d) $[-10^{-7}, 10^{-7}]$ by $[-0.1, 0.3]$

FIGURE 5

EXAMPLE 4 | Guess the value of $\lim_{x \rightarrow 0} \frac{\sin x}{x}$.

SOLUTION The function $f(x) = (\sin x)/x$ is not defined when $x = 0$. Using a calculator (and remembering that $\sin x$ means the sine of the angle whose *radian* measure is x), we construct a table of values correct to eight decimal places. From the table at the left and the graph in Figure 6 we guess that

x	$\frac{\sin x}{x}$
± 1.0	0.84147098
± 0.5	0.95885108
± 0.4	0.97354586
± 0.3	0.98506736
± 0.2	0.99334665
± 0.1	0.99833417
± 0.05	0.99958339
± 0.01	0.99998333
± 0.005	0.99999583
± 0.001	0.99999983

$$\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$$

This guess is in fact correct, as will be proved in the next section using a geometric argument.

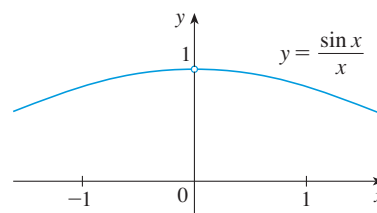


FIGURE 6

EXAMPLE 5 | Investigate $\lim_{x \rightarrow 0} \sin \frac{\pi}{x}$.

SOLUTION Again the function $f(x) = \sin(\pi/x)$ is undefined at 0. Evaluating the function for some small values of x , we get

$$f(1) = \sin \pi = 0 \qquad f\left(\frac{1}{2}\right) = \sin 2\pi = 0$$

$$f\left(\frac{1}{3}\right) = \sin 3\pi = 0 \qquad f\left(\frac{1}{4}\right) = \sin 4\pi = 0$$

$$f(0.1) = \sin 10\pi = 0 \qquad f(0.01) = \sin 100\pi = 0$$

Similarly, $f(0.001) = f(0.0001) = 0$. On the basis of this information we might be tempted to guess that

$$\lim_{x \rightarrow 0} \sin \frac{\pi}{x} = 0$$

but this time our **guess is wrong**. Note that although $f(1/n) = \sin n\pi = 0$ for any integer n , it is also true that $f(x) = 1$ for infinitely many values of x that approach 0. You can see this from the graph of f shown in Figure 7.

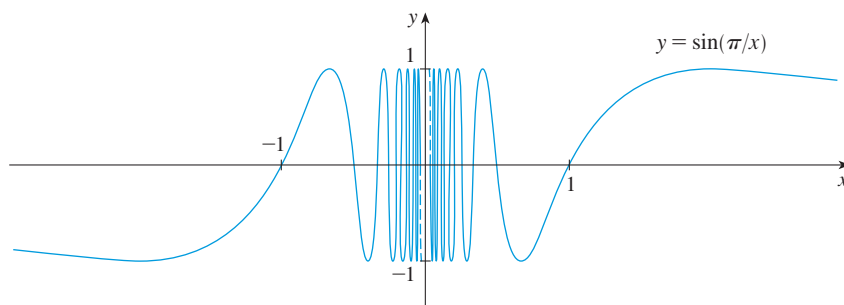


FIGURE 7

The dashed lines near the y -axis indicate that the values of $\sin(\pi/x)$ oscillate between 1 and -1 infinitely often as x approaches 0. (See Exercise 43.) Since the values of $f(x)$ do not approach a fixed number as x approaches 0,

$$\lim_{x \rightarrow 0} \sin \frac{\pi}{x} \text{ does not exist}$$

x	$x^3 + \frac{\cos 5x}{10,000}$
1	1.000028
0.5	0.124920
0.1	0.001088
0.05	0.000222
0.01	0.000101

EXAMPLE 6 | Find $\lim_{x \rightarrow 0} \left(x^3 + \frac{\cos 5x}{10,000} \right)$.

SOLUTION As before, we construct a table of values. From the first table in the margin it appears that

$$\lim_{x \rightarrow 0} \left(x^3 + \frac{\cos 5x}{10,000} \right) = 0$$

But if we persevere with smaller values of x , the second table suggests that

$$\lim_{x \rightarrow 0} \left(x^3 + \frac{\cos 5x}{10,000} \right) = 0.000100 = \frac{1}{10,000}$$

Later we will see that $\lim_{x \rightarrow 0} \cos 5x = 1$; then it follows that the limit is 0.0001.

x	$x^3 + \frac{\cos 5x}{10,000}$
0.005	0.00010009
0.001	0.00010000



Examples 5 and 6 illustrate some of **the pitfalls in guessing the value of a limit**. It is easy to guess the wrong value if we use inappropriate values of x , but it is difficult to know when to stop calculating values. And, as the discussion after Example 3 shows, sometimes calculators and computers give the wrong values. In the next section, however, we will develop foolproof methods for calculating limits.

EXAMPLE 7 | **BB** **Catastrophic population collapse** In later chapters we will explore a model for population size in which catastrophic collapse and extinction occurs if the habitat is degraded beyond a critical level. (See Examples 4.1.6 and 7.2.3 and Exercise 7.2.20.) We will see that the population size is given by the function

$$N(K) = \begin{cases} 0 & \text{if } 0 \leq K < a \\ K & \text{if } K \geq a \end{cases}$$

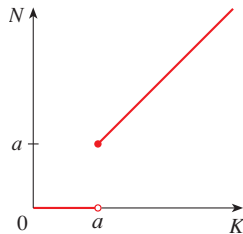


FIGURE 8

where a is a positive number and K is a nonnegative constant that measures habitat quality. (It is referred to as the carrying capacity. See Figure 8.)

As K approaches a from the left, $N(K)$ approaches 0. As K approaches a from the right, $N(K)$ approaches a . There is no single number that $N(K)$ approaches as K approaches a . Therefore $\lim_{K \rightarrow a} N(K)$ does not exist. ■

■ One-Sided Limits

We noticed in Example 7 that $N(K)$ approaches 0 as K approaches a from the left and $N(K)$ approaches a as K approaches a from the right. We indicate this situation symbolically by writing

$$\lim_{K \rightarrow a^-} N(K) = 0 \quad \text{and} \quad \lim_{K \rightarrow a^+} N(K) = a$$

The notation $K \rightarrow a^-$ indicates that we consider only values of K that are less than a . Likewise, $K \rightarrow a^+$ indicates that we consider only values of K that are greater than a .

(2) Definition We write

$$\lim_{x \rightarrow a^-} f(x) = L$$

and say the **left-hand limit of $f(x)$ as x approaches a** [or the **limit of $f(x)$ as x approaches a from the left**] is equal to L if we can make the values of $f(x)$ arbitrarily close to L by taking x to be sufficiently close to a and x less than a .

Notice that Definition 2 differs from Definition 1 only in that we require x to be less than a . Similarly, if we require that x be greater than a , we get “the **right-hand limit of $f(x)$ as x approaches a** is equal to L ” and we write

$$\lim_{x \rightarrow a^+} f(x) = L$$

Thus the notation $x \rightarrow a^+$ means that we consider only $x > a$. These definitions are illustrated in Figure 9.

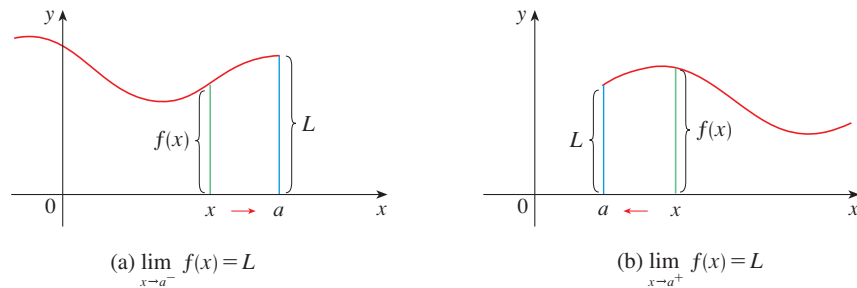


FIGURE 9

By comparing Definition 1 with the definitions of one-sided limits, we see that the following is true.

$$(3) \quad \lim_{x \rightarrow a} f(x) = L \quad \text{if and only if} \quad \lim_{x \rightarrow a^-} f(x) = L \quad \text{and} \quad \lim_{x \rightarrow a^+} f(x) = L$$



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EXAMPLE 8 | Bird population American robins breed in early spring and pairs produce a clutch of three to five eggs, laid one day apart. Only about 25% of chicks survive their first year. The graph in Figure 10 shows the population size $P(t)$ of a flock of robins for a 7-day period in spring, where t is the number of days starting at noon on day 0.

- (a) What happened at $t = 2$? At $t = 3$? Between $t = 4$ and $t = 5$?
 (b) What are the values of the following limits (if they exist)?

$$\lim_{t \rightarrow 2^-} P(t) \quad \lim_{t \rightarrow 2^+} P(t) \quad \lim_{t \rightarrow 2} P(t) \quad \lim_{t \rightarrow 6} P(t)$$

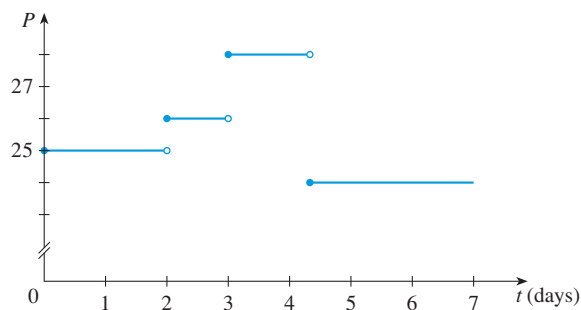


FIGURE 10

SOLUTION

- (a) We see from Figure 10 that at $t = 2$ the robin population increased by one, so the most likely explanation is that a chick hatched. At $t = 3$ two more chicks hatched. Between $t = 4$ and $t = 5$ the robin population decreased by 4, so four of the robins died, perhaps because of a predator.

- (b) We read the following values from the graph in Figure 10:

$$\lim_{t \rightarrow 2^-} P(t) = 25 \quad \lim_{t \rightarrow 2^+} P(t) = 26$$

Because these limits are different, we conclude that

$$\lim_{t \rightarrow 2} P(t) \text{ does not exist}$$

On the other hand,

$$\lim_{t \rightarrow 6^-} P(t) = 24 = \lim_{t \rightarrow 6^+} P(t)$$

so

$$\lim_{t \rightarrow 6} P(t) = 24$$

Infinite Limits

EXAMPLE 9 Find $\lim_{x \rightarrow 0} \frac{1}{x^2}$ if it exists.

SOLUTION As x becomes close to 0, x^2 also becomes close to 0, and $1/x^2$ becomes very large. (See the table in the margin.) In fact, it appears from the graph of the function $f(x) = 1/x^2$ shown in Figure 11 that the values of $f(x)$ can be made arbitrarily large by taking x close enough to 0. Thus the values of $f(x)$ do not approach a number, so $\lim_{x \rightarrow 0} (1/x^2)$ does not exist.

To indicate the kind of behavior exhibited in Example 9, we use the notation

$$\lim_{x \rightarrow 0} \frac{1}{x^2} = \infty$$



This does not mean that we are regarding ∞ as a number. Nor does it mean that the limit exists. It simply expresses the particular way in which the limit does not exist: $1/x^2$ can be made as large as we like by taking x close enough to 0.

In general, we write symbolically

$$\lim_{x \rightarrow a} f(x) = \infty$$

to indicate that the values of $f(x)$ tend to become larger and larger (or “increase without bound”) as x becomes closer and closer to a .

x	$\frac{1}{x^2}$
± 1	1
± 0.5	4
± 0.2	25
± 0.1	100
± 0.05	400
± 0.01	10,000
± 0.001	1,000,000

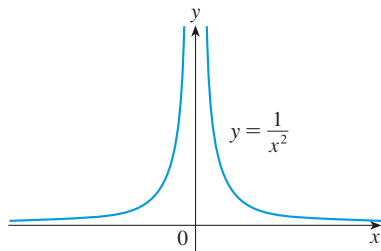


FIGURE 11

(4) Definition Let f be a function defined on both sides of a , except possibly at a itself. Then

$$\lim_{x \rightarrow a} f(x) = \infty$$

means that the values of $f(x)$ can be made arbitrarily large (as large as we please) by taking x sufficiently close to a but not equal to a .

Another notation for $\lim_{x \rightarrow a} f(x) = \infty$ is

$$f(x) \rightarrow \infty \text{ as } x \rightarrow a$$

Again, the symbol ∞ is not a number, but the expression $\lim_{x \rightarrow a} f(x) = \infty$ is often read as

“the limit of $f(x)$, as x approaches a , is infinity”

or

“ $f(x)$ becomes infinite as x approaches a ”

or

“ $f(x)$ increases without bound as x approaches a ”

This definition is illustrated graphically in Figure 12.

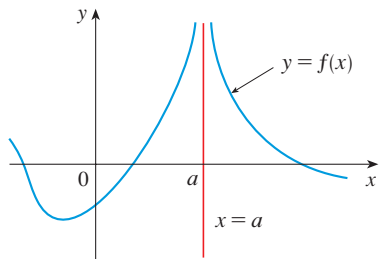


FIGURE 12

$$\lim_{x \rightarrow a} f(x) = \infty$$

When we say a number is “large negative,” we mean that it is negative but its magnitude (absolute value) is large.

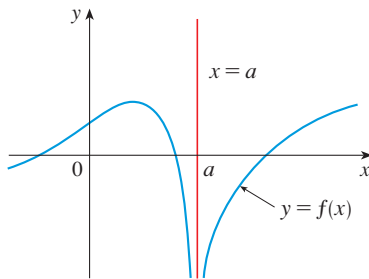


FIGURE 13
 $\lim_{x \rightarrow a} f(x) = -\infty$

A similar sort of limit, for functions that become large negative as x gets close to a , is defined in Definition 5 and is illustrated in Figure 13.

(5) Definition Let f be defined on both sides of a , except possibly at a itself. Then

$$\lim_{x \rightarrow a} f(x) = -\infty$$

means that the values of $f(x)$ can be made arbitrarily large negative by taking x sufficiently close to a but not equal to a .

The symbol $\lim_{x \rightarrow a} f(x) = -\infty$ can be read as “the limit of $f(x)$, as x approaches a , is negative infinity” or “ $f(x)$ decreases without bound as x approaches a .” As an example we have

$$\lim_{x \rightarrow 0} \left(-\frac{1}{x^2} \right) = -\infty$$

Similar definitions can be given for the one-sided infinite limits

$$\lim_{x \rightarrow a^-} f(x) = \infty$$

$$\lim_{x \rightarrow a^+} f(x) = \infty$$

$$\lim_{x \rightarrow a^-} f(x) = -\infty$$

$$\lim_{x \rightarrow a^+} f(x) = -\infty$$

remembering that $x \rightarrow a^-$ means that we consider only values of x that are less than a , and similarly $x \rightarrow a^+$ means that we consider only $x > a$. Illustrations of these four cases are given in Figure 14.

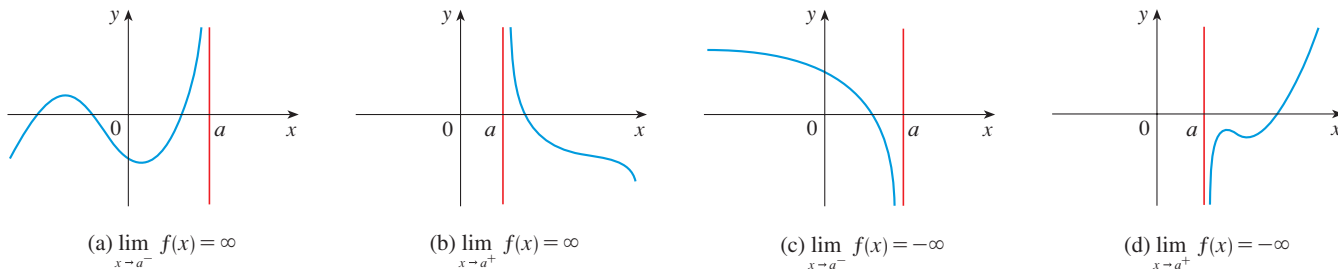


FIGURE 14

(6) Definition The line $x = a$ is called a **vertical asymptote** of the curve $y = f(x)$ if at least one of the following statements is true:

$$\lim_{x \rightarrow a} f(x) = \infty$$

$$\lim_{x \rightarrow a^-} f(x) = \infty$$

$$\lim_{x \rightarrow a^+} f(x) = \infty$$

$$\lim_{x \rightarrow a} f(x) = -\infty$$

$$\lim_{x \rightarrow a^-} f(x) = -\infty$$

$$\lim_{x \rightarrow a^+} f(x) = -\infty$$

For instance, the y -axis is a vertical asymptote of the curve $y = 1/x^2$ because $\lim_{x \rightarrow 0} (1/x^2) = \infty$. In Figure 14 the line $x = a$ is a vertical asymptote in each of the four cases shown. In general, knowledge of vertical asymptotes is very useful in sketching graphs.

EXAMPLE 10 | **BB** Anesthesiology In Example 1.2.5 we discussed the use by anesthesiologists of ventilators during surgery. We saw that

$$C = \frac{P}{V}$$

where C is the steady state concentration of CO_2 in the lungs and V is the ventilation rate. As $V \rightarrow 0^+$, $C \rightarrow \infty$ when P is constant and so

$$\lim_{V \rightarrow 0^+} \frac{P}{V} = \infty$$

Therefore in Figure 15 the C -axis is a vertical asymptote of the graph of C . This means that as the ventilation rate becomes very low, the concentration of carbon dioxide in the lungs becomes very high. (Note: This does not mean that, in reality, we expect the CO_2 level in the lungs to become arbitrarily high as the ventilation rate approaches 0. Remember, this is simply a *model* of how CO_2 concentration depends on ventilation rate. It is therefore a simplification of reality. Mathematical models often fail to be accurate descriptions of reality under extreme conditions, and this is an example.)

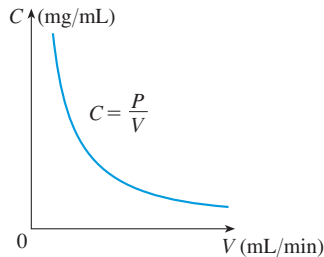


FIGURE 15

EXAMPLE 11 | Find $\lim_{x \rightarrow 3^+} \frac{2x}{x-3}$ and $\lim_{x \rightarrow 3^-} \frac{2x}{x-3}$.

SOLUTION If x is close to 3 but larger than 3, then the denominator $x - 3$ is a small positive number and $2x$ is close to 6. So the quotient $2x/(x - 3)$ is a large *positive* number. Thus, intuitively, we see that

$$\lim_{x \rightarrow 3^+} \frac{2x}{x-3} = \infty$$

Likewise, if x is close to 3 but smaller than 3, then $x - 3$ is a small negative number but $2x$ is still a positive number (close to 6). So $2x/(x - 3)$ is a numerically large *negative* number. Thus

$$\lim_{x \rightarrow 3^-} \frac{2x}{x-3} = -\infty$$

The graph of the curve $y = 2x/(x - 3)$ is given in Figure 16. The line $x = 3$ is a vertical asymptote.

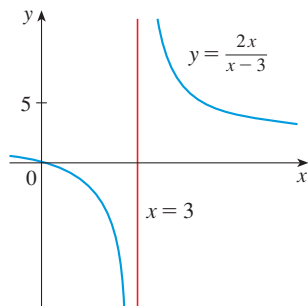


FIGURE 16

EXAMPLE 12 | Find the vertical asymptotes of $f(x) = \tan x$.

SOLUTION Because

$$\tan x = \frac{\sin x}{\cos x}$$

there are potential vertical asymptotes where $\cos x = 0$. In fact, since $\cos x \rightarrow 0^+$ as $x \rightarrow (\pi/2)^-$ and $\cos x \rightarrow 0^-$ as $x \rightarrow (\pi/2)^+$, whereas $\sin x$ is positive when x is near $\pi/2$, we have

$$\lim_{x \rightarrow (\pi/2)^-} \tan x = \infty \quad \text{and} \quad \lim_{x \rightarrow (\pi/2)^+} \tan x = -\infty$$

This shows that the line $x = \pi/2$ is a vertical asymptote. Similar reasoning shows that the lines $x = (2n + 1)\pi/2$, where n is an integer, are all vertical asymptotes of $f(x) = \tan x$. The graph in Figure 17 confirms this.

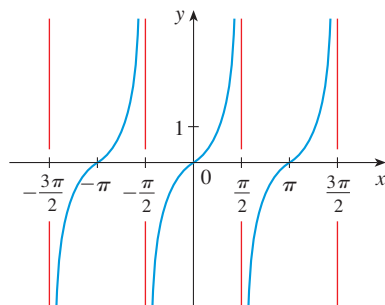


FIGURE 17
 $y = \tan x$

Another example of a function whose graph has a vertical asymptote is the natural logarithmic function $y = \ln x$. From Figure 18 we see that

$$\lim_{x \rightarrow 0^+} \ln x = -\infty$$

and so the line $x = 0$ (the y -axis) is a vertical asymptote. In fact, the same is true for $y = \log_b x$ provided that $b > 1$. (See Figures 11 and 12 in Section 1.5.)

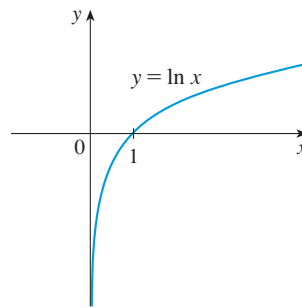


FIGURE 18

The y -axis is a vertical asymptote of the natural logarithmic function.

EXERCISES 2.3

1. Explain in your own words what is meant by the equation

$$\lim_{x \rightarrow 2} f(x) = 5$$

Is it possible for this statement to be true and yet $f(2) = 3$? Explain.

2. Explain what it means to say that

$$\lim_{x \rightarrow 1^-} f(x) = 3 \quad \text{and} \quad \lim_{x \rightarrow 1^+} f(x) = 7$$

In this situation is it possible that $\lim_{x \rightarrow 1} f(x)$ exists? Explain.

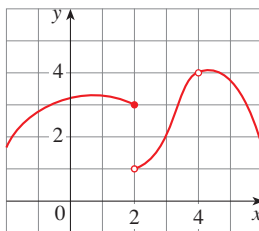
3. Explain the meaning of each of the following.

$$(a) \lim_{x \rightarrow -3} f(x) = \infty \quad (b) \lim_{x \rightarrow 4^+} f(x) = -\infty$$

4. Use the given graph of f to state the value of each quantity, if it exists. If it does not exist, explain why.

$$(a) \lim_{x \rightarrow 2^-} f(x) \quad (b) \lim_{x \rightarrow 2^+} f(x) \quad (c) \lim_{x \rightarrow 2} f(x)$$

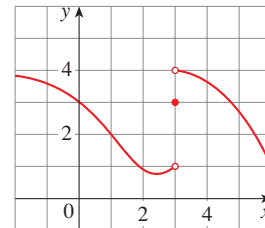
$$(d) f(2) \quad (e) \lim_{x \rightarrow 4} f(x) \quad (f) f(4)$$



5. For the function f whose graph is given, state the value of each quantity, if it exists. If it does not exist, explain why.

$$(a) \lim_{x \rightarrow 1} f(x) \quad (b) \lim_{x \rightarrow 3^-} f(x) \quad (c) \lim_{x \rightarrow 3^+} f(x)$$

$$(d) \lim_{x \rightarrow 3} f(x) \quad (e) f(3)$$



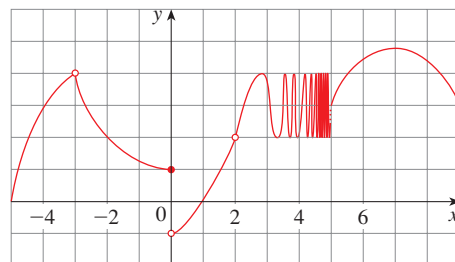
6. For the function h whose graph is given, state the value of each quantity, if it exists. If it does not exist, explain why.

$$(a) \lim_{x \rightarrow -3^-} h(x) \quad (b) \lim_{x \rightarrow -3^+} h(x) \quad (c) \lim_{x \rightarrow -3} h(x)$$

$$(d) h(-3) \quad (e) \lim_{x \rightarrow 0^-} h(x) \quad (f) \lim_{x \rightarrow 0^+} h(x)$$

$$(g) \lim_{x \rightarrow 0} h(x) \quad (h) h(0) \quad (i) \lim_{x \rightarrow 2} h(x)$$

$$(j) h(2) \quad (k) \lim_{x \rightarrow 5^+} h(x) \quad (l) \lim_{x \rightarrow 5^-} h(x)$$



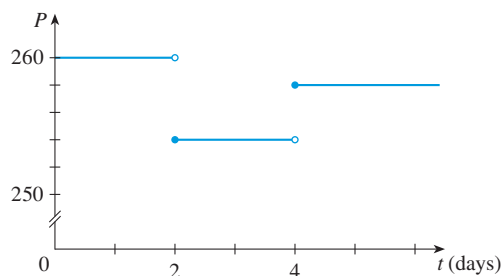
7. The **population of a village** is $P(t)$, t days after June 1. Use the graph of P to state the value of each limit, if it exists. If it doesn't exist, explain why.

(a) $\lim_{t \rightarrow 2^-} P(t)$ (b) $\lim_{t \rightarrow 2^+} P(t)$ (c) $\lim_{t \rightarrow 2} P(t)$

(d) $\lim_{t \rightarrow 4^-} P(t)$ (e) $\lim_{t \rightarrow 4^+} P(t)$ (f) $\lim_{t \rightarrow 4} P(t)$

(g) $\lim_{x \rightarrow 5} P(t)$

- (h) What do you think happened on June 3rd and 5th?



8. For the function R whose graph is shown, state the following.

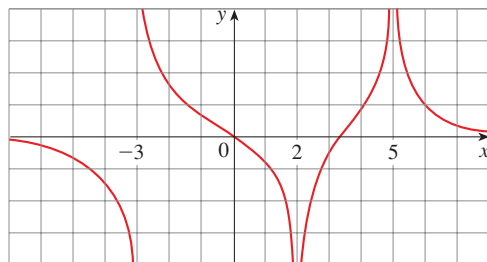
(a) $\lim_{x \rightarrow 2} R(x)$

(b) $\lim_{x \rightarrow 5} R(x)$

(c) $\lim_{x \rightarrow -3^-} R(x)$

(d) $\lim_{x \rightarrow -3^+} R(x)$

- (e) The equations of the vertical asymptotes.



9. For the function g whose graph is given, state the following.

(a) $\lim_{x \rightarrow 0} g(x)$

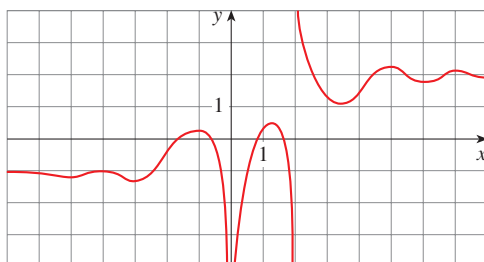
(b) $\lim_{x \rightarrow 2^-} g(x)$

(c) $\lim_{x \rightarrow 2^+} g(x)$

(d) $\lim_{x \rightarrow \infty} g(x)$

(e) $\lim_{x \rightarrow -\infty} g(x)$

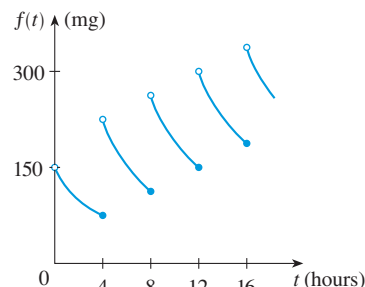
- (f) The equations of the asymptotes



10. **Drug injections** A patient receives a 150-mg injection of a drug every 4 hours. The graph shows the amount $f(t)$ of the drug in the bloodstream after t hours. Find

$$\lim_{x \rightarrow 12^-} f(t) \quad \text{and} \quad \lim_{x \rightarrow 12^+} f(t)$$

and explain the significance of these one-sided limits.



- 11–18 Sketch the graph of an example of a function f that satisfies all of the given conditions.

11. $\lim_{x \rightarrow 3^+} f(x) = 4$, $\lim_{x \rightarrow 3^-} f(x) = 2$, $\lim_{x \rightarrow -2} f(x) = 2$,
 $f(3) = 3$, $f(-2) = 1$

12. $\lim_{x \rightarrow 0^-} f(x) = 2$, $\lim_{x \rightarrow 0^+} f(x) = 0$, $\lim_{x \rightarrow 4^-} f(x) = 3$,
 $\lim_{x \rightarrow 4^+} f(x) = 0$, $f(0) = 2$, $f(4) = 1$

13. $\lim_{x \rightarrow 0} f(x) = -\infty$, $\lim_{x \rightarrow -\infty} f(x) = 5$, $\lim_{x \rightarrow \infty} f(x) = -5$

14. $\lim_{x \rightarrow 2} f(x) = \infty$, $\lim_{x \rightarrow -2^+} f(x) = \infty$, $\lim_{x \rightarrow -2^-} f(x) = -\infty$,
 $\lim_{x \rightarrow -\infty} f(x) = 0$, $\lim_{x \rightarrow \infty} f(x) = 0$, $f(0) = 0$

15. $\lim_{x \rightarrow 2} f(x) = -\infty$, $\lim_{x \rightarrow 2} f(x) = \infty$, $\lim_{x \rightarrow -\infty} f(x) = 0$,
 $\lim_{x \rightarrow 0^+} f(x) = \infty$, $\lim_{x \rightarrow 0^-} f(x) = -\infty$

16. $\lim_{x \rightarrow \infty} f(x) = 3$, $\lim_{x \rightarrow 2^-} f(x) = \infty$, $\lim_{x \rightarrow 2^+} f(x) = -\infty$,
 f is odd

17. $f(0) = 3$, $\lim_{x \rightarrow 0^-} f(x) = 4$, $\lim_{x \rightarrow 0^+} f(x) = 2$,
 $\lim_{x \rightarrow -\infty} f(x) = -\infty$, $\lim_{x \rightarrow 4^-} f(x) = -\infty$, $\lim_{x \rightarrow 4^+} f(x) = \infty$,
 $\lim_{x \rightarrow \infty} f(x) = 3$

18. $\lim_{x \rightarrow \infty} f(x) = -\infty$, $\lim_{x \rightarrow -\infty} f(x) = 2$, $f(0) = 0$, f is even

- 19–22 Guess the value of the limit (if it exists) by evaluating the function at the given numbers (correct to six decimal places).

19. $\lim_{x \rightarrow 2} \frac{x^2 - 2x}{x^2 - x - x}$,

$x = 2.5, 2.1, 2.05, 2.01, 2.005, 2.001,$
 $1.9, 1.95, 1.99, 1.995, 1.999$

$$20. \lim_{x \rightarrow -1} \frac{x^2 - 2x}{x^2 - x - 2},$$

$x = 0, -0.5, -0.9, -0.95, -0.99, -0.999,$
 $-2, -1.5, -1.1, -1.01, -1.001$

$$21. \lim_{t \rightarrow 0} \frac{e^{5t} - 1}{t}, \quad t = \pm 0.5, \pm 0.1, \pm 0.01, \pm 0.001, \pm 0.0001$$

$$22. \lim_{h \rightarrow 0} \frac{(2+h)^5 - 32}{h},$$

$h = \pm 0.5, \pm 0.1, \pm 0.01, \pm 0.001, \pm 0.0001$

23–26 Use a table of values to estimate the value of the limit. If you have a graphing device, use it to confirm your result graphically.

$$23. \lim_{x \rightarrow 0} \frac{\sqrt{x+4} - 2}{x}$$

$$24. \lim_{x \rightarrow 0} \frac{\tan 3x}{\tan 5x}$$

$$25. \lim_{x \rightarrow 1} \frac{x^6 - 1}{x^{10} - 1}$$

$$26. \lim_{x \rightarrow 0} \frac{9^x - 5^x}{x}$$

-  **27.** (a) By graphing the function $f(x) = (\cos 2x - \cos x)/x^2$ and zooming in toward the point where the graph crosses the y -axis, estimate the value of $\lim_{x \rightarrow 0} f(x)$.
 (b) Check your answer in part (a) by evaluating $f(x)$ for values of x that approach 0.

-  **28.** (a) Estimate the value of

$$\lim_{x \rightarrow 0} \frac{\sin x}{\sin \pi x}$$

by graphing the function $f(x) = (\sin x)/(\sin \pi x)$. State your answer correct to two decimal places.

- (b) Check your answer in part (a) by evaluating $f(x)$ for values of x that approach 0.

29–37 Determine the infinite limit.

$$29. \lim_{x \rightarrow -3^+} \frac{x+2}{x+3}$$

$$30. \lim_{x \rightarrow -3^-} \frac{x+2}{x+3}$$

$$31. \lim_{x \rightarrow 1} \frac{2-x}{(x-1)^2}$$

$$32. \lim_{x \rightarrow 5^-} \frac{e^x}{(x-5)^3}$$

$$33. \lim_{x \rightarrow 3^+} \ln(x^2 - 9)$$

$$34. \lim_{x \rightarrow \pi^-} \cot x$$

$$35. \lim_{x \rightarrow 2\pi^-} x \csc x$$

$$36. \lim_{x \rightarrow 2^-} \frac{x^2 - 2x}{x^2 - 4x + 4}$$

$$37. \lim_{x \rightarrow 2^+} \frac{x^2 - 2x - 8}{x^2 - 5x + 6}$$

- 38.** (a) Find the vertical asymptotes of the function

$$y = \frac{x^2 + 1}{3x - 2x^2}$$

-  (b) Confirm your answer to part (a) by graphing the function.

39. Determine $\lim_{x \rightarrow 1^-} \frac{1}{x^3 - 1}$ and $\lim_{x \rightarrow 1^+} \frac{1}{x^3 - 1}$

- (a) by evaluating $f(x) = 1/(x^3 - 1)$ for values of x that approach 1 from the left and from the right,
 (b) by reasoning as in Example 11, and
 (c) from a graph of f .

-  **40.** (a) Graph the function $f(x) = e^x + \ln|x - 4|$ for $0 \leq x \leq 5$. Do you think the graph is an accurate representation of f ?

- (b) How would you get a graph that represents f better?

- 41.** (a) Estimate the value of the limit $\lim_{x \rightarrow 0} (1+x)^{1/x}$ to five decimal places. Does this number look familiar?

-  (b) Illustrate part (a) by graphing the function $y = (1+x)^{1/x}$.

- 42.** (a) Evaluate the function $f(x) = x^2 - (2^x/1000)$ for $x = 1, 0.8, 0.6, 0.4, 0.2, 0.1$, and 0.05 , and guess the value of

$$\lim_{x \rightarrow 0} \left(x^2 - \frac{2^x}{1000} \right)$$

- (b) Evaluate $f(x)$ for $x = 0.04, 0.02, 0.01, 0.005, 0.003$, and 0.001 . Guess again.

-  **43.** Graph the function $f(x) = \sin(\pi/x)$ of Example 5 in the viewing rectangle $[-1, 1]$ by $[-1, 1]$. Then zoom in toward the origin several times. Comment on the behavior of this function.

- 44.** In the theory of relativity, the mass of a particle with velocity v is

$$m = \frac{m_0}{\sqrt{1 - v^2/c^2}}$$

where m_0 is the mass of the particle at rest and c is the speed of light. What happens as $v \rightarrow c^-$?

2.4 Limits: Algebraic Methods

In Section 2.3 we used calculators and graphs to guess the values of limits, but we saw that such methods don't always lead to the correct answer. In this section we use the following properties of limits, called the *Limit Laws*, to calculate limits.

■ The Limit Laws

The following five properties of limits are similar to the ones we have seen before, but some of the other properties of limits apply only when the variable x approaches a finite number.

Limit Laws for Functions

Suppose that c is a constant and the limits

$$\lim_{x \rightarrow a} f(x) \quad \text{and} \quad \lim_{x \rightarrow a} g(x)$$

exist. Then

1. $\lim_{x \rightarrow a} [f(x) + g(x)] = \lim_{x \rightarrow a} f(x) + \lim_{x \rightarrow a} g(x)$
2. $\lim_{x \rightarrow a} [f(x) - g(x)] = \lim_{x \rightarrow a} f(x) - \lim_{x \rightarrow a} g(x)$
3. $\lim_{x \rightarrow a} [cf(x)] = c \lim_{x \rightarrow a} f(x)$
4. $\lim_{x \rightarrow a} [f(x)g(x)] = \lim_{x \rightarrow a} f(x) \cdot \lim_{x \rightarrow a} g(x)$
5. $\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \frac{\lim_{x \rightarrow a} f(x)}{\lim_{x \rightarrow a} g(x)}$ if $\lim_{x \rightarrow a} g(x) \neq 0$

These laws can be stated verbally as follows:

- | | |
|------------------------------|---|
| Sum Law | 1. The limit of a sum is the sum of the limits. |
| Difference Law | 2. The limit of a difference is the difference of the limits. |
| Constant Multiple Law | 3. The limit of a constant times a function is the constant times the limit of the function. |
| Product Law | 4. The limit of a product is the product of the limits |
| Quotient Law | 5. The limit of a quotient is the quotient of the limits (provided that the limit of the denominator is not 0). |

It is easy to believe that these properties are true. For instance, if $f(x)$ is close to L and $g(x)$ is close to M , it is reasonable to conclude that $f(x) + g(x)$ is close to $L + M$. This gives us an intuitive basis for believing that Law 1 is true. All of these laws can be proved using the precise definition of a limit. In Appendix E we give the proof of Law 1.

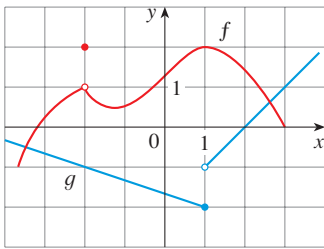


FIGURE 1

EXAMPLE 1 | Use the Limit Laws and the graphs of f and g in Figure 1 to evaluate the following limits, if they exist.

$$(a) \lim_{x \rightarrow -2} [f(x) + 5g(x)] \quad (b) \lim_{x \rightarrow 1} [f(x)g(x)] \quad (c) \lim_{x \rightarrow 2} \frac{f(x)}{g(x)}$$

SOLUTION

(a) From the graphs of f and g we see that

$$\lim_{x \rightarrow -2} f(x) = 1 \quad \text{and} \quad \lim_{x \rightarrow -2} g(x) = -1$$

Therefore we have

$$\begin{aligned} \lim_{x \rightarrow -2} [f(x) + 5g(x)] &= \lim_{x \rightarrow -2} f(x) + \lim_{x \rightarrow -2} [5g(x)] && \text{(by Law 1)} \\ &= \lim_{x \rightarrow -2} f(x) + 5 \lim_{x \rightarrow -2} g(x) && \text{(by Law 3)} \\ &= 1 + 5(-1) = -4 \end{aligned}$$

(b) We see that $\lim_{x \rightarrow 1} f(x) = 2$. But $\lim_{x \rightarrow 1} g(x)$ does not exist because the left and right limits are different:

$$\lim_{x \rightarrow 1^-} g(x) = -2 \quad \lim_{x \rightarrow 1^+} g(x) = -1$$

So we can't use Law 4 for the desired limit. But we can use Law 4 for the one-sided limits:

$$\lim_{x \rightarrow 1^-} [f(x)g(x)] = 2 \cdot (-2) = -4 \quad \lim_{x \rightarrow 1^+} [f(x)g(x)] = 2 \cdot (-1) = -2$$

The left and right limits aren't equal, so $\lim_{x \rightarrow 1} [f(x)g(x)]$ does not exist.

(c) The graphs show that

$$\lim_{x \rightarrow 2} f(x) \approx 1.4 \quad \text{and} \quad \lim_{x \rightarrow 2} g(x) = 0$$

Because the limit of the denominator is 0, we can't use Law 5. The given limit does not exist because the denominator approaches 0 while the numerator approaches a nonzero number. ■

If we use the Product Law repeatedly with $g(x) = f(x)$, we obtain the following law.

Power Law

$$6. \lim_{x \rightarrow a} [f(x)]^n = \left[\lim_{x \rightarrow a} f(x) \right]^n \quad \text{where } n \text{ is a positive integer}$$

In applying these six limit laws, we need to use two special limits:

$$7. \lim_{x \rightarrow a} c = c$$

$$8. \lim_{x \rightarrow a} x = a$$

These limits are obvious from an intuitive point of view (state them in words or draw graphs of $y = c$ and $y = x$).

If we now put $f(x) = x$ in Law 6 and use Law 8, we get another useful special limit.

$$9. \lim_{x \rightarrow a} x^n = a^n \quad \text{where } n \text{ is a positive integer}$$

A similar limit holds for roots as follows.

$$10. \lim_{x \rightarrow a} \sqrt[n]{x} = \sqrt[n]{a} \quad \text{where } n \text{ is a positive integer}$$

(If n is even, we assume that $a > 0$.)

More generally, we have the following law.

Root Law

$$11. \lim_{x \rightarrow a} \sqrt[n]{f(x)} = \sqrt[n]{\lim_{x \rightarrow a} f(x)} \quad \text{where } n \text{ is a positive integer}$$

[If n is even, we assume that $\lim_{x \rightarrow a} f(x) > 0$.]

EXAMPLE 2 | Evaluate the following limits and justify each step.

$$(a) \lim_{x \rightarrow 5} (2x^2 - 3x + 4) \qquad (b) \lim_{x \rightarrow -2} \frac{x^3 + 2x^2 - 1}{5 - 3x}$$

SOLUTION

$$\begin{aligned} (a) \quad \lim_{x \rightarrow 5} (2x^2 - 3x + 4) &= \lim_{x \rightarrow 5} (2x^2) - \lim_{x \rightarrow 5} (3x) + \lim_{x \rightarrow 5} 4 && \text{(by Laws 2 and 1)} \\ &= 2 \lim_{x \rightarrow 5} x^2 - 3 \lim_{x \rightarrow 5} x + \lim_{x \rightarrow 5} 4 && \text{(by 3)} \\ &= 2(5^2) - 3(5) + 4 && \text{(by 9, 8, and 7)} \\ &= 39 \end{aligned}$$

(b) We start by using Law 5, but its use is fully justified only at the final stage when we see that the limits of the numerator and denominator exist and the limit of the denominator is not 0.

$$\begin{aligned} \lim_{x \rightarrow -2} \frac{x^3 + 2x^2 - 1}{5 - 3x} &= \frac{\lim_{x \rightarrow -2} (x^3 + 2x^2 - 1)}{\lim_{x \rightarrow -2} (5 - 3x)} && \text{(by Law 5)} \\ &= \frac{\lim_{x \rightarrow -2} x^3 + 2 \lim_{x \rightarrow -2} x^2 - \lim_{x \rightarrow -2} 1}{\lim_{x \rightarrow -2} 5 - 3 \lim_{x \rightarrow -2} x} && \text{(by 1, 2, and 3)} \\ &= \frac{(-2)^3 + 2(-2)^2 - 1}{5 - 3(-2)} && \text{(by 9, 8, and 7)} \\ &= -\frac{1}{11} \end{aligned}$$

Newton and Limits

Isaac Newton was born on Christmas Day in 1642, the year of Galileo's death. When he entered Cambridge University in 1661 Newton didn't know much mathematics, but he learned quickly by reading Euclid and Descartes and by attending the lectures of Isaac Barrow. Cambridge was closed because of the plague in 1665 and 1666, and Newton returned home to reflect on what he had learned. Those two years were amazingly productive for at that time he made four of his major discoveries: (1) his representation of functions as sums of infinite series, including the binomial theorem; (2) his work on differential and integral calculus; (3) his laws of motion and law of universal gravitation; and (4) his prism experiments on the nature of light and color. Because of a fear of controversy and criticism, he was reluctant to publish his discoveries and it wasn't until 1687, at the urging of the astronomer Halley, that Newton published *Principia Mathematica*. In this work, the greatest scientific treatise ever written, Newton set forth his version of calculus and used it to investigate mechanics, fluid dynamics, and wave motion, and to explain the motion of planets and comets.

The beginnings of calculus are found in the calculations of areas and volumes by ancient Greek scholars such as Eudoxus and Archimedes. Although aspects of the idea of a limit are implicit in their "method of exhaustion," Eudoxus and Archimedes never explicitly formulated the concept of a limit. Likewise, mathematicians such as Cavalieri, Fermat, and Barrow, the immediate precursors of Newton in the development of calculus, did not actually use limits. It was Isaac Newton who was the first to talk explicitly about limits. He explained that the main idea behind limits is that quantities "approach nearer than by any given difference." Newton stated that the limit was the basic concept in calculus, but it was left to later mathematicians like Cauchy to clarify his ideas about limits.

NOTE If we let $f(x) = 2x^2 - 3x + 4$, then $f(5) = 39$. In other words, we would have gotten the correct answer in Example 2(a) by substituting 5 for x . Similarly, direct substitution provides the correct answer in part (b). The functions in Example 2 are a polynomial and a rational function, respectively, and similar use of the Limit Laws proves that direct substitution always works for such functions (see Exercise 45). We state this fact as follows.

Direct Substitution Property If f is a polynomial or a rational function and a is in the domain of f , then

$$\lim_{x \rightarrow a} f(x) = f(a)$$

Functions with the Direct Substitution Property are called *continuous at a* and will be studied in Section 2.5. However, not all limits can be evaluated by direct substitution, as the following examples show.

EXAMPLE 3 | Find $\lim_{x \rightarrow 1} \frac{x^2 - 1}{x - 1}$.

SOLUTION Let $f(x) = (x^2 - 1)/(x - 1)$. We can't find the limit by substituting $x = 1$ because $f(1)$ isn't defined. Nor can we apply the Quotient Law, because the limit of the denominator is 0. Instead, we need to do some preliminary algebra. We factor the numerator as a difference of squares:

$$\frac{x^2 - 1}{x - 1} = \frac{(x - 1)(x + 1)}{x - 1}$$

The numerator and denominator have a common factor of $x - 1$. When we take the limit as x approaches 1, we have $x \neq 1$ and so $x - 1 \neq 0$. Therefore we can cancel the common factor and compute the limit as follows:

$$\lim_{x \rightarrow 1} \frac{x^2 - 1}{x - 1} = \lim_{x \rightarrow 1} \frac{(x - 1)(x + 1)}{x - 1} = \lim_{x \rightarrow 1} (x + 1) = 1 + 1 = 2$$

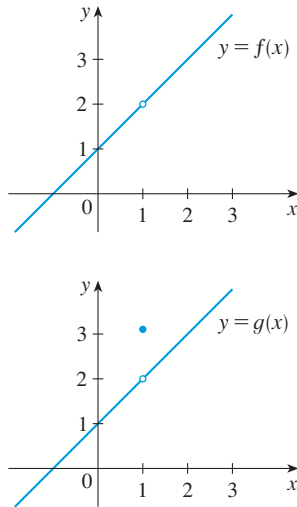
NOTE In Example 3 we were able to compute the limit by replacing the given function $f(x) = (x^2 - 1)/(x - 1)$ by a simpler function, $g(x) = x + 1$, with the same limit. This is valid because $f(x) = g(x)$ except when $x = 1$, and in computing a limit as x approaches 1 we don't consider what happens when x is actually *equal* to 1. In general, we have the following useful fact.

If $f(x) = g(x)$ when $x \neq a$, then $\lim_{x \rightarrow a} f(x) = \lim_{x \rightarrow a} g(x)$, provided the limits exist.

EXAMPLE 4 | Find $\lim_{x \rightarrow 1} g(x)$ where

$$g(x) = \begin{cases} x + 1 & \text{if } x \neq 1 \\ \pi & \text{if } x = 1 \end{cases}$$

SOLUTION Here g is defined at $x = 1$ and $g(1) = \pi$, but the value of a limit as x approaches 1 does not depend on the value of the function at 1. Since $g(x) = x + 1$ for

**FIGURE 2**

The graphs of the functions f (from Example 3) and g (from Example 4)

$x \neq 1$, we have

$$\lim_{x \rightarrow 1} g(x) = \lim_{x \rightarrow 1} (x + 1) = 2$$

Note that the values of the functions in Examples 3 and 4 are identical except when $x = 1$ (see Figure 2) and so they have the same limit as x approaches 1.

EXAMPLE 5 | Evaluate $\lim_{h \rightarrow 0} \frac{(3 + h)^2 - 9}{h}$.

SOLUTION If we define

$$F(h) = \frac{(3 + h)^2 - 9}{h}$$

then, as in Example 3, we can't compute $\lim_{h \rightarrow 0} F(h)$ by letting $h = 0$ since $F(0)$ is undefined. But if we simplify $F(h)$ algebraically, we find that

$$F(h) = \frac{(9 + 6h + h^2) - 9}{h} = \frac{6h + h^2}{h} = 6 + h$$

(Recall that we consider only $h \neq 0$ when letting h approach 0.) Thus

$$\lim_{h \rightarrow 0} \frac{(3 + h)^2 - 9}{h} = \lim_{h \rightarrow 0} (6 + h) = 6$$

EXAMPLE 6 | Find $\lim_{t \rightarrow 0} \frac{\sqrt{t^2 + 9} - 3}{t^2}$.

SOLUTION We can't apply the Quotient Law immediately, since the limit of the denominator is 0. Here the preliminary algebra consists of rationalizing the numerator:

$$\begin{aligned} \lim_{t \rightarrow 0} \frac{\sqrt{t^2 + 9} - 3}{t^2} &= \lim_{t \rightarrow 0} \frac{\sqrt{t^2 + 9} - 3}{t^2} \cdot \frac{\sqrt{t^2 + 9} + 3}{\sqrt{t^2 + 9} + 3} \\ &= \lim_{t \rightarrow 0} \frac{(t^2 + 9) - 9}{t^2(\sqrt{t^2 + 9} + 3)} \\ &= \lim_{t \rightarrow 0} \frac{t^2}{t^2(\sqrt{t^2 + 9} + 3)} \\ &= \lim_{t \rightarrow 0} \frac{1}{\sqrt{t^2 + 9} + 3} \\ &= \frac{1}{\sqrt{\lim_{t \rightarrow 0} (t^2 + 9)} + 3} \\ &= \frac{1}{3 + 3} = \frac{1}{6} \end{aligned}$$

This calculation confirms the guess that we made in Example 2.3.3.

EXAMPLE 7 | HIV prevalence and incidence The number of people with HIV in New York in the early 1980s is well described by the function $P(t) = 80 + 16t + 121t^2$, where t measures the years since 1982. (The graph of P is shown in Figure 3.) Epidemiologists call $P(t)$ the *prevalence* of HIV at time t . The *incidence* of HIV is the number of new infections per unit time over a specified time interval.

- What was the HIV incidence during the 12 months of 1983?
- What was the HIV incidence during the first month of 1983?
- Calculate the limiting value of the incidence as the time interval shrinks to zero.

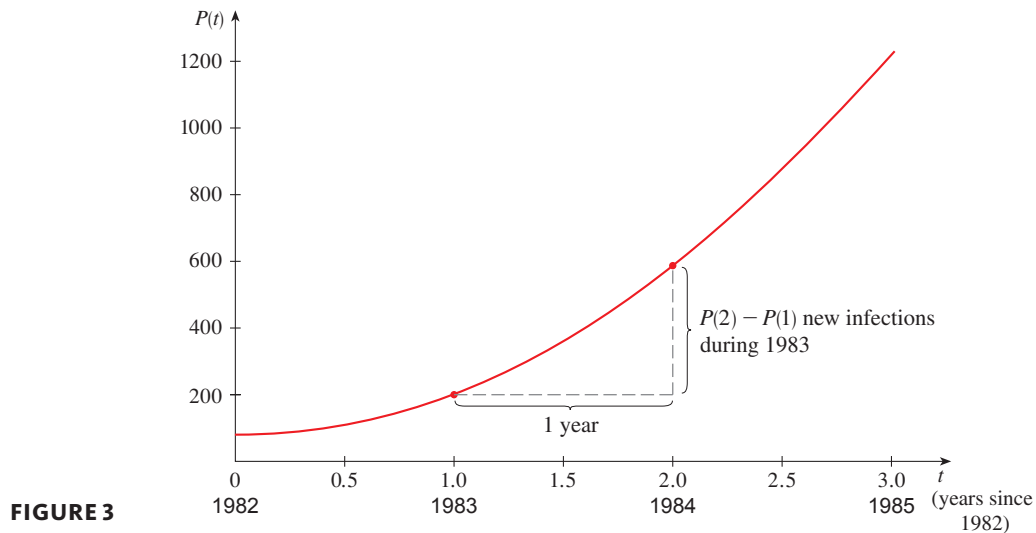


FIGURE 3

SOLUTION The number of infections in 1983 was

$$P(1) = 80 + 16(1) + 121(1^2) = 217$$

- (a) The HIV incidence during the 12 months of 1983 was

$$\frac{P(2) - P(1)}{1} = \frac{80 + 16(2) + 121(2^2) - 217}{1} = 379 \text{ infections/year}$$

- (b) The HIV incidence during the first month of 1983 was

$$\frac{P(1\frac{1}{12}) - P(1)}{\frac{1}{12}} = \frac{\frac{34,465}{144} - 217}{\frac{1}{12}} = \frac{3217}{12} \approx 268 \text{ infections/year}$$

- (c) The limiting value of the incidence is

$$\begin{aligned} \lim_{h \rightarrow 0} \frac{P(1+h) - P(1)}{h} &= \lim_{h \rightarrow 0} \frac{80 + 16(1+h) + 121(1+h)^2 - 217}{h} \\ &= \lim_{h \rightarrow 0} \frac{217 + 258h + 121h^2 - 217}{h} \\ &= \lim_{h \rightarrow 0} (258 + 121h) = 258 \text{ infections/year} \end{aligned}$$

Additional Properties of Limits

Some limits are best calculated by first finding the left- and right-hand limits. The following theorem is a reminder of what we discovered in Section 2.3. It says that a two-sided limit exists if and only if both of the one-sided limits exist and are equal.

$$(1) \text{ Theorem } \lim_{x \rightarrow a} f(x) = L \quad \text{if and only if} \quad \lim_{x \rightarrow a^-} f(x) = L = \lim_{x \rightarrow a^+} f(x)$$

When computing one-sided limits, we use the fact that the Limit Laws also hold for one-sided limits.

EXAMPLE 8 | Show that $\lim_{x \rightarrow 0} |x| = 0$.

SOLUTION Recall that

$$|x| = \begin{cases} x & \text{if } x \geq 0 \\ -x & \text{if } x < 0 \end{cases}$$

Since $|x| = x$ for $x > 0$, we have

$$\lim_{x \rightarrow 0^+} |x| = \lim_{x \rightarrow 0^+} x = 0$$

For $x < 0$ we have $|x| = -x$ and so

$$\lim_{x \rightarrow 0^-} |x| = \lim_{x \rightarrow 0^-} (-x) = 0$$

Therefore, by Theorem 1,

$$\lim_{x \rightarrow 0} |x| = 0$$

The result of Example 8 looks plausible from Figure 4.

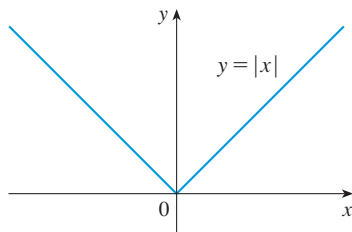


FIGURE 4

EXAMPLE 9 | Prove that $\lim_{x \rightarrow 0} \frac{|x|}{x}$ does not exist.

SOLUTION

$$\lim_{x \rightarrow 0^+} \frac{|x|}{x} = \lim_{x \rightarrow 0^+} \frac{x}{x} = \lim_{x \rightarrow 0^+} 1 = 1$$

$$\lim_{x \rightarrow 0^-} \frac{|x|}{x} = \lim_{x \rightarrow 0^-} \frac{-x}{x} = \lim_{x \rightarrow 0^-} (-1) = -1$$

Since the right- and left-hand limits are different, it follows from Theorem 1 that $\lim_{x \rightarrow 0} |x|/x$ does not exist. The graph of the function $f(x) = |x|/x$ is shown in Figure 5 and supports the one-sided limits that we found.

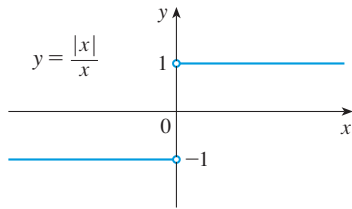


FIGURE 5

The next two theorems give two additional properties of limits. Both can be proved using the precise definition in Appendix D.

(2) Theorem If $f(x) \leq g(x)$ when x is near a (except possibly at a) and the limits of f and g both exist as x approaches a , then

$$\lim_{x \rightarrow a} f(x) \leq \lim_{x \rightarrow a} g(x)$$

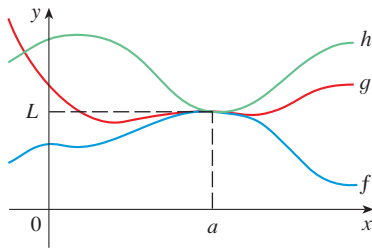


FIGURE 6

(3) The Squeeze Theorem If $f(x) \leq g(x) \leq h(x)$ when x is near a (except possibly at a) and

$$\lim_{x \rightarrow a} f(x) = \lim_{x \rightarrow a} h(x) = L$$

then

$$\lim_{x \rightarrow a} g(x) = L$$

The Squeeze Theorem, which is sometimes called the Sandwich Theorem or the Pinching Theorem, is illustrated by Figure 6. It says that if $g(x)$ is squeezed between $f(x)$ and $h(x)$ near a , and if f and h have the same limit L at a , then g is forced to have the same limit L at a .

EXAMPLE 10 | Show that $\lim_{x \rightarrow 0} x^2 \sin \frac{1}{x} = 0$.

SOLUTION First note that we **cannot** use



$$\lim_{x \rightarrow 0} x^2 \sin \frac{1}{x} = \lim_{x \rightarrow 0} x^2 \cdot \lim_{x \rightarrow 0} \sin \frac{1}{x}$$

because $\lim_{x \rightarrow 0} \sin(1/x)$ does not exist (see Example 2.3.5).

Instead we apply the Squeeze Theorem, and so we need to find a function f smaller than $g(x) = x^2 \sin(1/x)$ and a function h bigger than g such that both $f(x)$ and $h(x)$ approach 0 as x approaches 0. To do this we use our knowledge of the sine function. Because the sine of any number lies between -1 and 1 , we can write

$$(4) \quad -1 \leq \sin \frac{1}{x} \leq 1$$

Any inequality remains true when multiplied by a positive number. We know that $x^2 \geq 0$ for all x and so, multiplying each side of the inequalities in (4) by x^2 , we get

$$-x^2 \leq x^2 \sin \frac{1}{x} \leq x^2$$

as illustrated by Figure 7. We know that

$$\lim_{x \rightarrow 0} x^2 = 0 \quad \text{and} \quad \lim_{x \rightarrow 0} (-x^2) = 0$$

Taking $f(x) = -x^2$, $g(x) = x^2 \sin(1/x)$, and $h(x) = x^2$ in the Squeeze Theorem, we obtain

$$\lim_{x \rightarrow 0} x^2 \sin \frac{1}{x} = 0$$

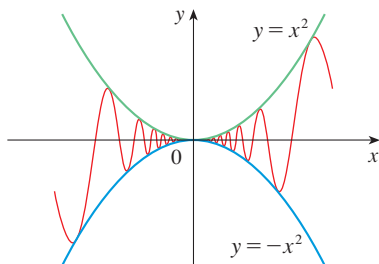


FIGURE 7

$$y = x^2 \sin(1/x)$$

■ Limits of Trigonometric Functions

We've seen that polynomials and rational functions satisfy the Direct Substitution Property. We will now show that the sine and cosine functions also satisfy that property. We know from the definitions of $\sin \theta$ and $\cos \theta$ that the coordinates of the point P in

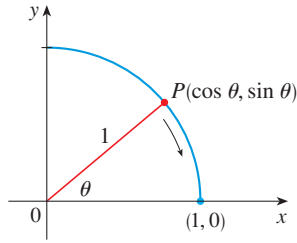


FIGURE 8

Another way to establish the limits in (5) is to use the inequality $\sin \theta < \theta$ (for $\theta > 0$), which is proved after Example 11.

Figure 8 are $(\cos \theta, \sin \theta)$. As $\theta \rightarrow 0$, we see that P approaches the point $(1, 0)$ and so $\cos \theta \rightarrow 1$ and $\sin \theta \rightarrow 0$. Thus

$$(5) \quad \lim_{\theta \rightarrow 0} \cos \theta = 1 \quad \lim_{\theta \rightarrow 0} \sin \theta = 0$$

Since $\cos 0 = 1$ and $\sin 0 = 0$, the equations in (5) assert that the cosine and sine functions satisfy the Direct Substitution Property at 0. The addition formulas for cosine and sine can then be used to deduce that these functions satisfy the Direct Substitution Property everywhere (see Exercises 47 and 48). In other words, for any real number a ,

$$\lim_{\theta \rightarrow a} \sin \theta = \sin a \quad \lim_{\theta \rightarrow a} \cos \theta = \cos a$$

This enables us to evaluate certain limits quite simply, as the next example shows.

EXAMPLE 11 | Evaluate $\lim_{x \rightarrow \pi} x \cos x$.

SOLUTION Using Limit Law 4 and the fact that the cosine function satisfies the Direct Substitution Property, we get

$$\begin{aligned} \lim_{x \rightarrow \pi} x \cos x &= \left(\lim_{x \rightarrow \pi} x \right) \left(\lim_{x \rightarrow \pi} \cos x \right) \\ &= \pi \cdot \cos \pi = -\pi \end{aligned}$$

In Example 2.3.4 we made the guess, on the basis of numerical and graphical evidence, that

$$(6) \quad \lim_{\theta \rightarrow 0} \frac{\sin \theta}{\theta} = 1$$

We can prove Equation 6 with help from the Squeeze Theorem. Assume first that θ lies between 0 and $\pi/2$. Figure 9(a) shows a sector of a circle with center O , central angle θ , and radius 1. BC is drawn perpendicular to OA . By the definition of radian measure, we have $\text{arc } AB = \theta$. Also $|BC| = |OB| \sin \theta = \sin \theta$. From the diagram we see that

$$|BC| < |AB| < \text{arc } AB$$

Therefore $\sin \theta < \theta$ so $\frac{\sin \theta}{\theta} < 1$

Let the tangent lines at A and B intersect at E . You can see from Figure 9(b) that the circumference of a circle is smaller than the length of a circumscribed polygon, and so $\text{arc } AB < |AE| + |EB|$. Thus

$$\begin{aligned} \theta = \text{arc } AB &< |AE| + |EB| < |AE| + |ED| \\ &= |AD| = |OA| \tan \theta = \tan \theta \end{aligned}$$

Therefore we have

$$\theta < \frac{\sin \theta}{\cos \theta} \quad \text{and so} \quad \cos \theta < \frac{\sin \theta}{\theta} < 1$$

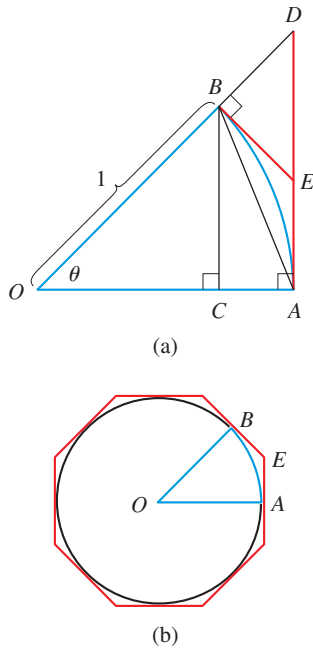


FIGURE 9

We know that $\lim_{\theta \rightarrow 0} 1 = 1$ and $\lim_{\theta \rightarrow 0} \cos \theta = 1$, so by the Squeeze Theorem, we have

$$\lim_{\theta \rightarrow 0^+} \frac{\sin \theta}{\theta} = 1$$

But the function $(\sin \theta)/\theta$ is an even function, so its right and left limits must be equal. Hence, we have

$$\lim_{\theta \rightarrow 0} \frac{\sin \theta}{\theta} = 1$$

so we have proved Equation 6.

EXAMPLE 12 | Find $\lim_{x \rightarrow 0} \frac{\sin 7x}{4x}$.

SOLUTION In order to apply Equation 6, we first rewrite the function by multiplying and dividing by 7:

Note that $\sin 7x \neq 7 \sin x$.

$$\frac{\sin 7x}{4x} = \frac{7}{4} \left(\frac{\sin 7x}{7x} \right)$$

Notice that as $x \rightarrow 0$, we have $7x \rightarrow 0$, and so, by Equation 6 with $\theta = 7x$,

$$\lim_{x \rightarrow 0} \frac{\sin 7x}{7x} = \lim_{7x \rightarrow 0} \frac{\sin(7x)}{7x} = 1$$

Thus

$$\begin{aligned} \lim_{x \rightarrow 0} \frac{\sin 7x}{4x} &= \lim_{x \rightarrow 0} \frac{7}{4} \left(\frac{\sin 7x}{7x} \right) \\ &= \frac{7}{4} \lim_{x \rightarrow 0} \frac{\sin 7x}{7x} = \frac{7}{4} \cdot 1 = \frac{7}{4} \end{aligned}$$

EXAMPLE 13 | Evaluate $\lim_{\theta \rightarrow 0} \frac{\cos \theta - 1}{\theta}$.

SOLUTION

We multiply numerator and denominator by $\cos \theta + 1$ in order to put the function in a form in which we can use the limits we know.

$$\begin{aligned} \lim_{\theta \rightarrow 0} \frac{\cos \theta - 1}{\theta} &= \lim_{\theta \rightarrow 0} \left(\frac{\cos \theta - 1}{\theta} \cdot \frac{\cos \theta + 1}{\cos \theta + 1} \right) = \lim_{\theta \rightarrow 0} \frac{\cos^2 \theta - 1}{\theta(\cos \theta + 1)} \\ &= \lim_{\theta \rightarrow 0} \frac{-\sin^2 \theta}{\theta(\cos \theta + 1)} = -\lim_{\theta \rightarrow 0} \left(\frac{\sin \theta}{\theta} \cdot \frac{\sin \theta}{\cos \theta + 1} \right) \\ &= -\lim_{\theta \rightarrow 0} \frac{\sin \theta}{\theta} \cdot \lim_{\theta \rightarrow 0} \frac{\sin \theta}{\cos \theta + 1} \\ &= -1 \cdot \left(\frac{0}{1 + 1} \right) = 0 \quad (\text{by Equation 6}) \end{aligned}$$

EXERCISES 2.4

1. Given that

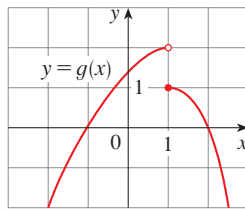
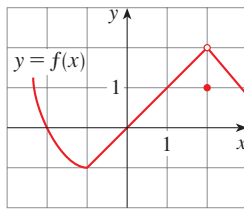
$$\lim_{x \rightarrow 2} f(x) = 4 \quad \lim_{x \rightarrow 2} g(x) = -2 \quad \lim_{x \rightarrow 2} h(x) = 0$$

find the limits that exist. If the limit does not exist, explain why.

$$\begin{array}{ll} \text{(a)} \lim_{x \rightarrow 2} [f(x) + 5g(x)] & \text{(b)} \lim_{x \rightarrow 2} [g(x)]^3 \\ \text{(c)} \lim_{x \rightarrow 2} \sqrt{f(x)} & \text{(d)} \lim_{x \rightarrow 2} \frac{3f(x)}{g(x)} \\ \text{(e)} \lim_{x \rightarrow 2} \frac{g(x)}{h(x)} & \text{(f)} \lim_{x \rightarrow 2} \frac{g(x)h(x)}{f(x)} \end{array}$$

 2. The graphs of f and g are given. Use them to evaluate each limit, if it exists. If the limit does not exist, explain why.

$$\begin{array}{ll} \text{(a)} \lim_{x \rightarrow 2} [f(x) + g(x)] & \text{(b)} \lim_{x \rightarrow 1} [f(x) + g(x)] \\ \text{(c)} \lim_{x \rightarrow 0} [f(x)g(x)] & \text{(d)} \lim_{x \rightarrow -1} \frac{f(x)}{g(x)} \\ \text{(e)} \lim_{x \rightarrow 2} [x^3 f(x)] & \text{(f)} \lim_{x \rightarrow 1} \sqrt{3 + f(x)} \end{array}$$



3–7 Evaluate the limit and justify each step by indicating the appropriate Limit Law(s).

$$3. \lim_{x \rightarrow -2} (3x^4 + 2x^2 - x + 1) \quad 4. \lim_{t \rightarrow -1} (t^2 + 1)^3(t + 3)^5$$

$$5. \lim_{x \rightarrow 2} \sqrt{\frac{2x^2 + 1}{3x - 2}} \quad 6. \lim_{x \rightarrow 0} \frac{\cos^4 x}{5 + 2x^3}$$

$$7. \lim_{\theta \rightarrow \pi/2} \theta \sin \theta$$

8. (a) What is wrong with the following equation?

$$\frac{x^2 + x - 6}{x - 2} = x + 3$$

(b) In view of part (a), explain why the equation

$$\lim_{x \rightarrow 2} \frac{x^2 + x - 6}{x - 2} = \lim_{x \rightarrow 2} (x + 3)$$

is correct.

9–24 Evaluate the limit, if it exists.

$$9. \lim_{x \rightarrow 5} \frac{x^2 - 6x + 5}{x - 5} \quad 10. \lim_{x \rightarrow 4} \frac{x^2 - 4x}{x^2 - 3x - 4}$$

$$11. \lim_{x \rightarrow 5} \frac{x^2 - 5x + 6}{x - 5}$$

$$12. \lim_{x \rightarrow -1} \frac{2x^2 + 3x + 1}{x^2 - 2x - 3}$$

$$13. \lim_{t \rightarrow -3} \frac{t^2 - 9}{2t^2 + 7t + 3}$$

$$14. \lim_{x \rightarrow -1} \frac{x^2 - 4x}{x^2 - 3x - 4}$$

$$15. \lim_{h \rightarrow 0} \frac{(4 + h)^2 - 16}{h}$$

$$16. \lim_{h \rightarrow 0} \frac{(2 + h)^3 - 8}{h}$$

$$17. \lim_{x \rightarrow -2} \frac{x + 2}{x^3 + 8}$$

$$18. \lim_{h \rightarrow 0} \frac{\sqrt{1 + h} - 1}{h}$$

$$19. \lim_{x \rightarrow -4} \frac{\frac{1}{4} + \frac{1}{x}}{4 + x}$$

$$20. \lim_{x \rightarrow -1} \frac{x^2 + 2x + 1}{x^4 - 1}$$

$$21. \lim_{x \rightarrow 16} \frac{4 - \sqrt{x}}{16x - x^2}$$

$$22. \lim_{t \rightarrow 0} \left(\frac{1}{t} - \frac{1}{t^2 + t} \right)$$

$$23. \lim_{t \rightarrow 0} \left(\frac{1}{t\sqrt{1+t}} - \frac{1}{t} \right)$$

$$24. \lim_{x \rightarrow -4} \frac{\sqrt{x^2 + 9} - 5}{x + 4}$$

25. (a) Estimate the value of

$$\lim_{x \rightarrow 0} \frac{x}{\sqrt{1 + 3x} - 1}$$

- by graphing the function $f(x) = x/(\sqrt{1 + 3x} - 1)$.
 (b) Make a table of values of $f(x)$ for x close to 0 and guess the value of the limit.
 (c) Use the Limit Laws to prove that your guess is correct.

26. (a) Use a graph of

$$f(x) = \frac{\sqrt{3 + x} - \sqrt{3}}{x}$$

- to estimate the value of $\lim_{x \rightarrow 0} f(x)$ to two decimal places.
 (b) Use a table of values of $f(x)$ to estimate the limit to four decimal places.
 (c) Use the Limit Laws to find the exact value of the limit.

 27. Use the Squeeze Theorem to show that $\lim_{x \rightarrow 0} (x^2 \cos 20\pi x) = 0$. Illustrate by graphing the functions $f(x) = -x^2$, $g(x) = x^2 \cos 20\pi x$, and $h(x) = x^2$ on the same screen.

28. Use the Squeeze Theorem to show that

$$\lim_{x \rightarrow 0} \sqrt{x^3 + x^2} \sin \frac{\pi}{x} = 0$$

 Illustrate by graphing the functions f , g , and h (in the notation of the Squeeze Theorem) on the same screen.

 29. If $4x - 9 \leq f(x) \leq x^2 - 4x + 7$ for $x \geq 0$, find $\lim_{x \rightarrow 4} f(x)$.

30. If $2x \leq g(x) \leq x^4 - x^2 + 2$ for all x , evaluate $\lim_{x \rightarrow 1} g(x)$.

31. Prove that $\lim_{x \rightarrow 0} x^4 \cos \frac{2}{x} = 0$.

32. **Gene regulation** Genes produce molecules called mRNA that go on to produce proteins. High concentrations of protein inhibit the production of mRNA, leading to stable gene regulation. This process has been modeled (see Section 10.3) to show that the concentration of mRNA over time is given by the equation

$$m(t) = \frac{1}{2}e^{-t}(\sin t - \cos t) + \frac{1}{2}$$

- (a) Evaluate $\lim_{t \rightarrow 0} m(t)$ and interpret your result.
 (b) Use the Squeeze Theorem to evaluate $\lim_{t \rightarrow \infty} m(t)$ and interpret your result.

33–36 Find the limit, if it exists. If the limit does not exist, explain why.

33. $\lim_{x \rightarrow 3} (2x + |x - 3|)$

34. $\lim_{x \rightarrow -6} \frac{2x + 12}{|x + 6|}$

35. $\lim_{x \rightarrow 0^-} \left(\frac{1}{x} - \frac{1}{|x|} \right)$

36. $\lim_{x \rightarrow -2} \frac{2 - |x|}{2 + x}$

37. Let $g(x) = \frac{x^2 + x - 6}{|x - 2|}$.

- (a) Find
 (i) $\lim_{x \rightarrow 2^+} g(x)$ (ii) $\lim_{x \rightarrow 2^-} g(x)$
 (b) Does $\lim_{x \rightarrow 2} g(x)$ exist?
 (c) Sketch the graph of g .

38. Let

$$f(x) = \begin{cases} x^2 + 1 & \text{if } x < 1 \\ (x - 2)^2 & \text{if } x \geq 1 \end{cases}$$

- (a) Find $\lim_{x \rightarrow 1^-} f(x)$ and $\lim_{x \rightarrow 1^+} f(x)$.
 (b) Does $\lim_{x \rightarrow 1} f(x)$ exist?
 (c) Sketch the graph of f .

39–44 Find the limit.

39. $\lim_{x \rightarrow 0} \frac{\sin 3x}{x}$

40. $\lim_{x \rightarrow 0} \frac{\sin 4x}{\sin 6x}$

41. $\lim_{t \rightarrow 0} \frac{\tan 6t}{\sin 2t}$

42. $\lim_{t \rightarrow 0} \frac{\sin^2 3t}{t^2}$

43. $\lim_{\theta \rightarrow 0} \frac{\sin \theta}{\theta + \tan \theta}$

44. $\lim_{x \rightarrow 0} x \cot x$

45. (a) If p is a polynomial, show that $\lim_{x \rightarrow a} p(x) = p(a)$.
 (b) If r is a rational function, use part (a) to show that $\lim_{x \rightarrow a} r(x) = r(a)$ for every number a in the domain of r .

46. In the theory of relativity, the Lorentz contraction formula

$$L = L_0 \sqrt{1 - v^2/c^2}$$

expresses the length L of an object as a function of its velocity v with respect to an observer, where L_0 is the length of the object at rest and c is the speed of light. Find $\lim_{v \rightarrow c^-} L$ and interpret the result. Why is a left-hand limit necessary?

47. To prove that sine has the Direct Substitution Property we need to show that $\lim_{x \rightarrow a} \sin x = \sin a$ for every real number a . If we let $h = x - a$, then $x = a + h$ and $x \rightarrow a \iff h \rightarrow 0$. So an equivalent statement is that

$$\lim_{h \rightarrow 0} \sin(a + h) = \sin a$$

Use (5) to show that this is true.

48. Prove that cosine has the Direct Substitution Property.

49. If $\lim_{x \rightarrow 1} \frac{f(x) - 8}{x - 1} = 10$, find $\lim_{x \rightarrow 1} f(x)$.

50. If $\lim_{x \rightarrow 0} \frac{f(x)}{x^2} = 5$, find the following limits.

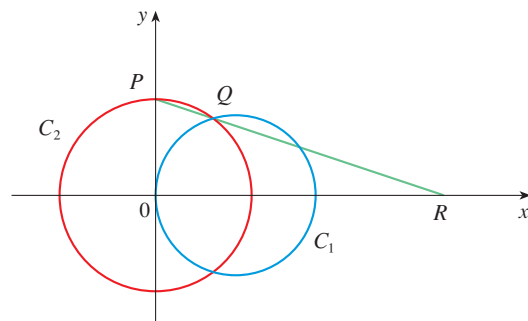
(a) $\lim_{x \rightarrow 0} f(x)$ (b) $\lim_{x \rightarrow 0} \frac{f(x)}{x}$

51. Is there a number a such that

$$\lim_{x \rightarrow -2} \frac{3x^2 + ax + a + 3}{x^2 + x - 2}$$

exists? If so, find the value of a and the value of the limit.

52. The figure shows a fixed circle C_1 with equation $(x - 1)^2 + y^2 = 1$ and a shrinking circle C_2 with radius r and center the origin. P is the point $(0, r)$, Q is the upper point of intersection of the two circles, and R is the point of intersection of the line PQ and the x -axis. What happens to R as C_2 shrinks, that is, as $r \rightarrow 0^+$?



2.5 Continuity

We noticed in Section 2.4 that the limit of a function as x approaches a can often be found simply by calculating the value of the function at a . Functions with this property are called *continuous at a* . We will see that the mathematical definition of continuity corresponds closely with the meaning of the word *continuity* in everyday language. (A continuous process is one that takes place gradually, without interruption or abrupt change.)

Definition of a Continuous Function

(1) Definition A function f is **continuous at a number a** if

$$\lim_{x \rightarrow a} f(x) = f(a)$$

As illustrated in Figure 1, if f is continuous, then the points $(x, f(x))$ on the graph of f approach the point $(a, f(a))$ on the graph. So there is no gap in the curve.

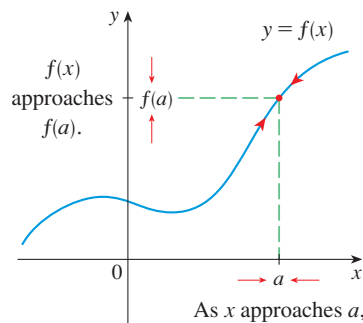


FIGURE 1

Notice that Definition 1 implicitly requires three things if f is continuous at a :

1. $f(a)$ is defined (that is, a is in the domain of f)
2. $\lim_{x \rightarrow a} f(x)$ exists
3. $\lim_{x \rightarrow a} f(x) = f(a)$

The definition says that f is continuous at a if $f(x)$ approaches $f(a)$ as x approaches a . Thus a continuous function f has the property that a small change in x produces only a small change in $f(x)$. In fact, the change in $f(x)$ can be kept as small as we please by keeping the change in x sufficiently small.

If f is defined near a (in other words, f is defined on an open interval containing a , except perhaps at a), we say that f is **discontinuous at a** (or f has a **discontinuity at a**) if f is not continuous at a .

Geometrically, you can think of a function that is continuous at every number in an interval as a function whose graph has no break in it. The graph can be drawn without removing your pen from the paper.

EXAMPLE 1 | Figure 2 shows the graph of a function f . At which numbers is f discontinuous? Why?

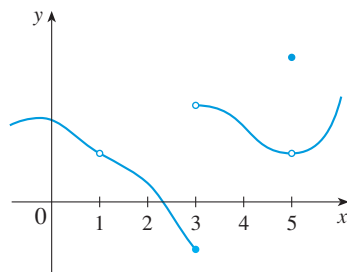


FIGURE 2

SOLUTION It looks as if there is a discontinuity when $a = 1$ because the graph has a break there. The official reason that f is discontinuous at 1 is that $f(1)$ is not defined.

The graph also has a break when $a = 3$, but the reason for the discontinuity is different. Here, $f(3)$ is defined, but $\lim_{x \rightarrow 3} f(x)$ does not exist (because the left and right limits are different). So f is discontinuous at 3.

What about $a = 5$? Here, $f(5)$ is defined and $\lim_{x \rightarrow 5} f(x)$ exists (because the left and right limits are the same). But

$$\lim_{x \rightarrow 5} f(x) \neq f(5)$$

So f is discontinuous at 5.

EXAMPLE 2 | Show that the function $f(x) = x^3 + 3x^2$ is continuous at every real number a .

SOLUTION Using the Limit Laws, we have

$$\begin{aligned}\lim_{x \rightarrow a} f(x) &= \lim_{x \rightarrow a} (x^3 + 3x^2) = \lim_{x \rightarrow a} x^3 + 3 \lim_{x \rightarrow a} x^2 \\ &= a^3 + 3a^2 = f(a)\end{aligned}$$

Therefore f is continuous at any number a . ■

EXAMPLE 3 | Where are each of the following functions discontinuous?

$$\begin{aligned}\text{(a)} \quad f(x) &= \frac{x^2 - x - 2}{x - 2} & \text{(b)} \quad f(x) &= \begin{cases} \frac{1}{x^2} & \text{if } x \neq 0 \\ 1 & \text{if } x = 0 \end{cases} \\ \text{(c)} \quad f(x) &= \begin{cases} \frac{x^2 - x - 2}{x - 2} & \text{if } x \neq 2 \\ 1 & \text{if } x = 2 \end{cases} & \text{(d)} \quad H(t) &= \begin{cases} 0 & \text{if } t < 0 \\ 1 & \text{if } t \geq 0 \end{cases}\end{aligned}$$

SOLUTION

(a) Notice that $f(2)$ is not defined, so f is discontinuous at 2. Later we'll see why f is continuous at all other numbers.

(b) Here $f(0) = 1$ is defined but

$$\lim_{x \rightarrow 0} f(x) = \lim_{x \rightarrow 0} \frac{1}{x^2}$$

does not exist. (See Example 2.3.9.) So f is discontinuous at 0.

(c) Here $f(2) = 1$ is defined and

$$\lim_{x \rightarrow 2} f(x) = \lim_{x \rightarrow 2} \frac{x^2 - x - 2}{x - 2} = \lim_{x \rightarrow 2} \frac{(x - 2)(x + 1)}{x - 2} = \lim_{x \rightarrow 2} (x + 1) = 3$$

exists. But

$$\lim_{x \rightarrow 2} f(x) \neq f(2)$$

so f is not continuous at 2.

(d) H is called the Heaviside function after the electrical engineer Oliver Heaviside (1853–1925) and can be used to describe an electric current that is turned on at time $t = 0$. Because

$$\lim_{t \rightarrow 0^-} H(t) = 0 \quad \text{and} \quad \lim_{t \rightarrow 0^+} H(t) = 1$$

we know that $\lim_{t \rightarrow 0} H(t)$ does not exist. So H is discontinuous at 0. ■

Figure 3 shows the graphs of the functions in Example 3. In each case the graph can't be drawn without lifting the pen from the paper because a hole or break or jump occurs in the graph. The kind of discontinuity illustrated in parts (a) and (c) is called **removable** because we could remove the discontinuity by redefining f at just the single number 2.

[The function $g(x) = x + 1$ is continuous.] The discontinuity in part (b) is called an **infinite discontinuity**. The discontinuity in part (d) is called a **jump discontinuity** because the function “jumps” from one value to another.

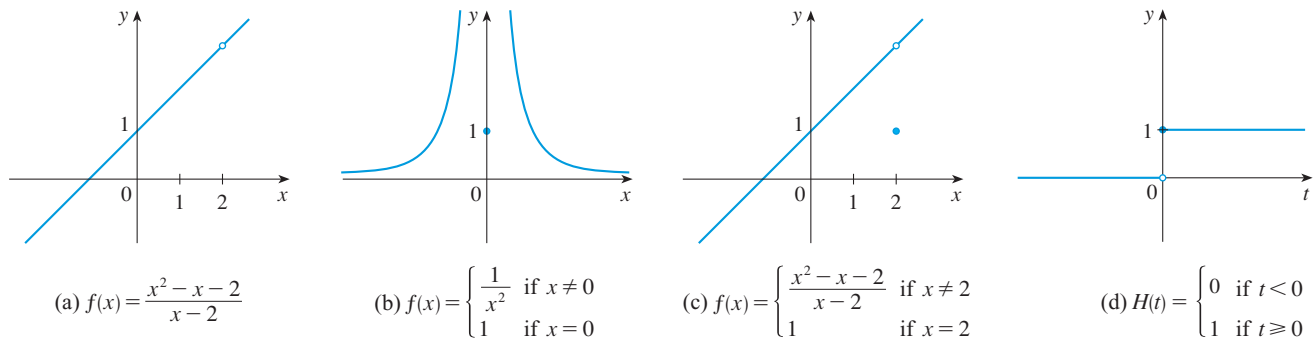


FIGURE 3

Graphs of the functions in Example 3

Physical phenomena are usually continuous. For instance, the displacement or velocity of a vehicle varies continuously with time, as does a person’s height. But discontinuities do occur in biology. Figure 4 shows the graph of the robin population from Example 2.3.8. The population function has a jump discontinuity whenever births or deaths occur, that is, at $t = 2, 3$, and 4.2 .

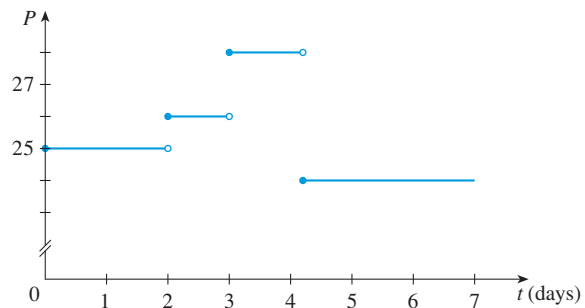


FIGURE 4



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EXAMPLE 4 | **BB** **Population harvesting and collapse** Intensive harvesting of a population, such as occurs for some fish species, can cause population extinction. How this extinction occurs, however, depends on the nature of the harvesting. For example, two of the models for harvesting in the project on page 438 give the following equations for the population size N (measured in thousands) as a function of harvesting effort h , where $h \geq 0$:

$$\text{Model 1} \quad N(h) = \begin{cases} 2(1 - h) & \text{if } h \leq 1 \\ 0 & \text{if } h > 1 \end{cases}$$

$$\text{Model 2} \quad N(h) = \begin{cases} \frac{1}{2}(1 + \sqrt{9 - 8h}) & \text{if } h \leq \frac{9}{8} \\ 0 & \text{if } h > \frac{9}{8} \end{cases}$$

Plot the population size for both models and comment on their continuity properties at the point where extinction occurs. What is the biological significance of the difference in continuity between the two models?

SOLUTION The graphs of both models are shown in Figure 5. In model 1 the function is continuous and a small increase in harvesting always causes a small decrease in population size. Thus, if the harvesting pressure is only increased by small amounts, we would be able to “see extinction coming.” But in model 2 a small increase in harvesting causes a discontinuous collapse in population size at $h = \frac{9}{8}$. In such cases a small increase in harvesting can cause the population to go extinct with no warning.

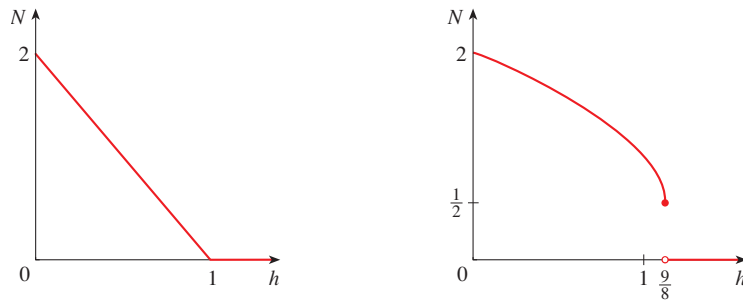


FIGURE 5

Model 1

Model 2

(2) Definition A function f is **continuous from the right at a number a** if

$$\lim_{x \rightarrow a^+} f(x) = f(a)$$

and f is **continuous from the left at a** if

$$\lim_{x \rightarrow a^-} f(x) = f(a)$$

Notice from Figure 4 that $P(t)$ is continuous from the right, but not from the left, at each of the jump discontinuities.

(3) Definition A function f is **continuous on an interval** if it is continuous at every number in the interval. (If f is defined only on one side of an endpoint of the interval, we understand *continuous* at the endpoint to mean *continuous from the right* or *continuous from the left*.)

Instead of using Definitions 1, 2, and 3 to verify the continuity of a function as we did in Example 2, it is often convenient to use the next theorem, which shows how to build complicated continuous functions from simple ones.

(4) Theorem If f and g are continuous at a and c is a constant, then the following functions are also continuous at a :

1. $f + g$

2. $f - g$

3. cf

4. fg

5. $\frac{f}{g}$ if $g(a) \neq 0$

PROOF Each of the five parts of this theorem follows from the corresponding Limit Law in Section 2.4. For instance, we give the proof of part 1. Since f and g are continuous at a , we have

$$\lim_{x \rightarrow a} f(x) = f(a) \quad \text{and} \quad \lim_{x \rightarrow a} g(x) = g(a)$$

Therefore

$$\begin{aligned} \lim_{x \rightarrow a} (f + g)(x) &= \lim_{x \rightarrow a} [f(x) + g(x)] \\ &= \lim_{x \rightarrow a} f(x) + \lim_{x \rightarrow a} g(x) \quad (\text{by Law 1}) \\ &= f(a) + g(a) \\ &= (f + g)(a) \end{aligned}$$

This shows that $f + g$ is continuous at a . ■

It follows from Theorem 4 and Definition 3 that if f and g are continuous on an interval, then so are the functions $f + g$, $f - g$, cf , fg , and (if g is never 0) f/g .

■ Which Functions Are Continuous?

It will be useful in our future work to know which functions are continuous and which are not. To that end let's begin to compile a list of functions that are known to be continuous. The following theorem was stated in Section 2.4 as the Direct Substitution Property.

(5) Theorem

- (a) Any polynomial is continuous everywhere; that is, it is continuous on $\mathbb{R} = (-\infty, \infty)$.
- (b) Any rational function is continuous wherever it is defined; that is, it is continuous on its domain.

PROOF

(a) A polynomial is a function of the form

$$P(x) = c_n x^n + c_{n-1} x^{n-1} + \cdots + c_1 x + c_0$$

where $c_0, c_1, \dots, c_n$ are constants. We know that

$$\lim_{x \rightarrow a} c_0 = c_0 \quad (\text{by Law 7})$$

$$\text{and} \quad \lim_{x \rightarrow a} x^m = a^m \quad m = 1, 2, \dots, n \quad (\text{by Law 9})$$

This equation is precisely the statement that the function $f(x) = x^m$ is a continuous function. Thus, by part 3 of Theorem 4, the function $g(x) = cx^m$ is continuous. Since P is a sum of functions of this form and a constant function, it follows from part 1 of Theorem 4 that P is continuous.

(b) A rational function is a function of the form

$$f(x) = \frac{P(x)}{Q(x)}$$

where P and Q are polynomials. The domain of f is $D = \{x \in \mathbb{R} \mid Q(x) \neq 0\}$. We know from part (a) that P and Q are continuous everywhere. Thus, by part 5 of Theorem 4, f is continuous at every number in D . ■

As an illustration of Theorem 5, observe that the volume of a sphere varies continuously with its radius because the formula $V(r) = \frac{4}{3}\pi r^3$ shows that V is a polynomial function of r . Likewise, if a ball is thrown vertically into the air with a velocity of 50 ft/s, then the height of the ball in feet t seconds later is given by the formula $h = 50t - 16t^2$. Again this is a polynomial function, so the height is a continuous function of the elapsed time.

Knowledge of which functions are continuous enables us to evaluate some limits very quickly, as the following example shows. Compare it with Example 2.4.2(b).

EXAMPLE 5 | Find $\lim_{x \rightarrow -2} \frac{x^3 + 2x^2 - 1}{5 - 3x}$.

SOLUTION The function

$$f(x) = \frac{x^3 + 2x^2 - 1}{5 - 3x}$$

is rational, so by Theorem 5 it is continuous on its domain, which is $\{x \mid x \neq \frac{5}{3}\}$. Therefore

$$\begin{aligned} \lim_{x \rightarrow -2} \frac{x^3 + 2x^2 - 1}{5 - 3x} &= \lim_{x \rightarrow -2} f(x) = f(-2) \\ &= \frac{(-2)^3 + 2(-2)^2 - 1}{5 - 3(-2)} = -\frac{1}{11} \end{aligned}$$

It turns out that most of the familiar functions are continuous at every number in their domains. For instance, Limit Law 10 (page 127) is exactly the statement that root functions are continuous.

From the appearance of the graphs of the sine and cosine functions (Figure 1.2.19), we would certainly guess that they are continuous. In fact we showed in Section 2.4 that they both satisfy the Direct Substitution Property, so indeed they are continuous everywhere.

It follows from part 5 of Theorem 4 that

$$\tan x = \frac{\sin x}{\cos x}$$

is continuous except where $\cos x = 0$. This happens when x is an odd integer multiple of $\pi/2$, so $y = \tan x$ has infinite discontinuities when $x = \pm\pi/2, \pm3\pi/2, \pm5\pi/2$, and so on (see Figure 6).

In Section 1.4 we defined the exponential function $y = b^x$ so as to fill in the holes in the graph of $y = b^x$ where x is rational. In other words, the very definition of $y = b^x$ makes it a continuous function on $\mathbb{R}$. The inverse function of any continuous one-to-one function is also continuous. (The graph of f^{-1} is obtained by reflecting the graph of f

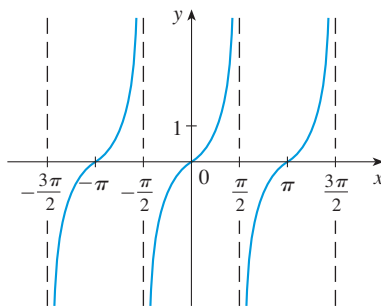


FIGURE 6
 $y = \tan x$

about the line $y = x$. So if the graph of f has no break in it, neither does the graph of f^{-1} .) Therefore the function $y = \log_b x$ is continuous on $(0, \infty)$ because it is the inverse function of $y = b^x$.

(6) Theorem The following types of functions are continuous at every number in their domains:

polynomials	rational functions
power and root functions	trigonometric functions
exponential functions	logarithmic functions

EXAMPLE 6 | Where is the function $f(x) = \frac{\ln x + e^x}{x^2 - 1}$ continuous?

SOLUTION We know from Theorem 6 that the function $y = \ln x$ is continuous for $x > 0$ and $y = e^x$ is continuous on $\mathbb{R}$. Thus, by part 1 of Theorem 4, $y = \ln x + e^x$ is continuous on $(0, \infty)$. The denominator, $y = x^2 - 1$, is a polynomial, so it is continuous everywhere. Therefore, by part 5 of Theorem 4, f is continuous at all positive numbers x except where $x^2 - 1 = 0$. So f is continuous on the intervals $(0, 1)$ and $(1, \infty)$. ■

EXAMPLE 7 | Evaluate $\lim_{x \rightarrow \pi} \frac{\sin x}{2 + \cos x}$.

SOLUTION Theorem 6 tells us that $y = \sin x$ is continuous. The function in the denominator, $y = 2 + \cos x$, is the sum of two continuous functions and is therefore continuous. Notice that this function is never 0 because $\cos x \geq -1$ for all x and so $2 + \cos x > 0$ everywhere. Thus the ratio

$$f(x) = \frac{\sin x}{2 + \cos x}$$

is continuous everywhere. Hence, by the definition of a continuous function,

$$\lim_{x \rightarrow \pi} \frac{\sin x}{2 + \cos x} = \lim_{x \rightarrow \pi} f(x) = f(\pi) = \frac{\sin \pi}{2 + \cos \pi} = \frac{0}{2 - 1} = 0$$

Another way of combining continuous functions f and g to get a new continuous function is to form the composite function $f \circ g$. This fact is a consequence of the following theorem.

This theorem says that a limit symbol can be moved through a function symbol if the function is continuous and the limit exists. In other words, the order of these two symbols can be reversed.

(7) Theorem If f is continuous at b and $\lim_{x \rightarrow a} g(x) = b$, then $\lim_{x \rightarrow a} f(g(x)) = f(b)$. In other words,

$$\lim_{x \rightarrow a} f(g(x)) = f\left(\lim_{x \rightarrow a} g(x)\right)$$

Intuitively, Theorem 7 is reasonable because if x is close to a , then $g(x)$ is close to b , and since f is continuous at b , if $g(x)$ is close to b , then $f(g(x))$ is close to $f(b)$.

(8) Theorem If g is continuous at a and f is continuous at $g(a)$, then the composite function $f \circ g$ given by $(f \circ g)(x) = f(g(x))$ is continuous at a .

This theorem is often expressed informally by saying “a continuous function of a continuous function is a continuous function.”

PROOF Since g is continuous at a , we have

$$\lim_{x \rightarrow a} g(x) = g(a)$$

Since f is continuous at $b = g(a)$, we can apply Theorem 7 to obtain

$$\lim_{x \rightarrow a} f(g(x)) = f(g(a))$$

which is precisely the statement that the function $h(x) = f(g(x))$ is continuous at a ; that is, $f \circ g$ is continuous at a . ■

EXAMPLE 8 | Where are the following functions continuous?

(a) $h(x) = \sin(x^2)$

(b) $F(x) = \ln(1 + \cos x)$

SOLUTION

(a) We have $h(x) = f(g(x))$, where

$$g(x) = x^2 \quad \text{and} \quad f(x) = \sin x$$

Now g is continuous on $\mathbb{R}$ since it is a polynomial, and f is also continuous everywhere. Thus $h = f \circ g$ is continuous on $\mathbb{R}$ by Theorem 8.

(b) We know from Theorem 6 that $f(x) = \ln x$ is continuous and $g(x) = 1 + \cos x$ is continuous (because both $y = 1$ and $y = \cos x$ are continuous). Therefore, by Theorem 8, $F(x) = f(g(x))$ is continuous wherever it is defined. Now $\ln(1 + \cos x)$ is defined when $1 + \cos x > 0$. So it is undefined when $\cos x = -1$, and this happens when $x = \pm\pi, \pm3\pi, \dots$. Thus F has discontinuities when x is an odd multiple of π and is continuous on the intervals between these values (see Figure 7). ■

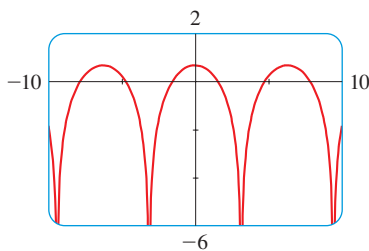


FIGURE 7
 $y = \ln(1 + \cos x)$

An important property of continuous functions is expressed by the following theorem, whose proof is found in more advanced books on calculus.

(9) The Intermediate Value Theorem Suppose that f is continuous on the closed interval $[a, b]$ and let N be any number between $f(a)$ and $f(b)$, where $f(a) \neq f(b)$. Then there exists a number c in (a, b) such that $f(c) = N$.

The Intermediate Value Theorem states that a continuous function takes on every intermediate value between the function values $f(a)$ and $f(b)$. It is illustrated by Figure 8. Note that the value N can be taken on once [as in part (a)] or more than once [as in part (b)].

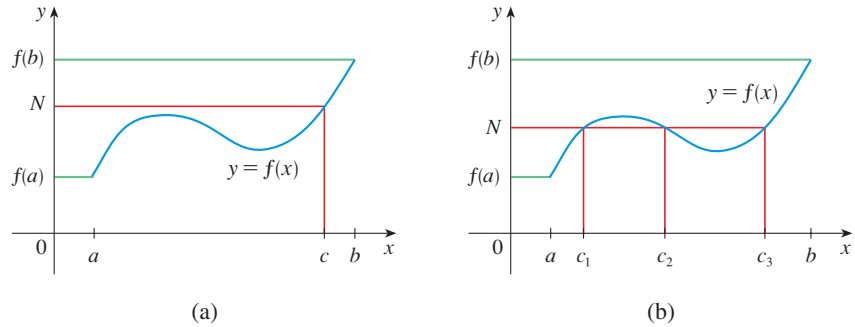


FIGURE 8

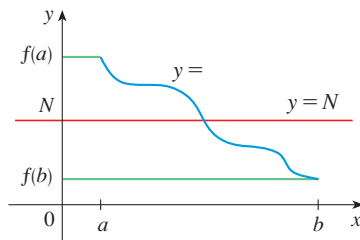


FIGURE 9

If we think of a continuous function as a function whose graph has no hole or break, then it is easy to believe that the Intermediate Value Theorem is true. In geometric terms it says that if any horizontal line $y = N$ is given between $y = f(a)$ and $y = f(b)$ as in Figure 9, then the graph of f can't jump over the line. It must intersect $y = N$ somewhere.

EXAMPLE 9 | BB Population harvesting and collapse (continued)

Model 1 in Example 4 for the size of a population as a function of harvesting rate is continuous for all harvesting rates, whereas model 2 has a jump discontinuity at the harvesting rate $h = \frac{9}{8}$. Provide a biological interpretation of the Intermediate Value Theorem in the context of these two models.

SOLUTION Since model 1 is continuous for all values of h , there is a harvesting rate that results in any population size between 2 and 0 (extinction). The discontinuity in model 2, however, means that there are some population sizes between 2 and 0 that cannot be obtained no matter what harvesting rate is used (for example, there is no harvesting rate h that results in a population size of $N = \frac{1}{4}$). This can have important conservation implications. For instance, suppose conservation biologists decide not to impose harvesting regulations unless the population has declined to a critical value of $N = \frac{1}{4}$. If model 1 is appropriate, then such regulations can prevent extinction. But if model 2 is appropriate, then the critical population size can never be reached no matter what the harvesting rate. Instead, the population will undergo irreversible collapse as harvesting rates increase before the population ever declines to the critical level. ■

One use of the Intermediate Value Theorem is in locating roots of equations, as shown in the following example.

EXAMPLE 10 | Show that there is a root of the equation

$$4x^3 - 6x^2 + 3x - 2 = 0$$

between 1 and 2.

SOLUTION Let $f(x) = 4x^3 - 6x^2 + 3x - 2$. We are looking for a solution of the given equation, that is, a number c between 1 and 2 such that $f(c) = 0$. Therefore we take $a = 1$, $b = 2$, and $N = 0$ in Theorem 9. We have

$$f(1) = 4 - 6 + 3 - 2 = -1 < 0$$

and

$$f(2) = 32 - 24 + 6 - 2 = 12 > 0$$

Thus $f(1) < 0 < f(2)$; that is, $N = 0$ is a number between $f(1)$ and $f(2)$. Now f is

continuous since it is a polynomial, so the Intermediate Value Theorem says there is a number c between 1 and 2 such that $f(c) = 0$. In other words, the equation $4x^3 - 6x^2 + 3x - 2 = 0$ has at least one root c in the interval $(1, 2)$.

In fact, we can locate a root more precisely by using the Intermediate Value Theorem again. Since

$$f(1.2) = -0.128 < 0 \quad \text{and} \quad f(1.3) = 0.548 > 0$$

a root must lie between 1.2 and 1.3. A calculator gives, by trial and error,

$$f(1.22) = -0.007008 < 0 \quad \text{and} \quad f(1.23) = 0.056068 > 0$$

so a root lies in the interval $(1.22, 1.23)$. ■

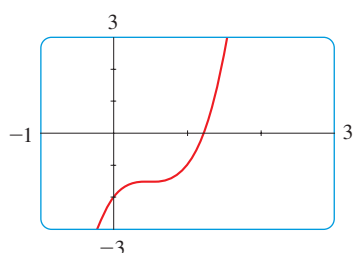


FIGURE 10

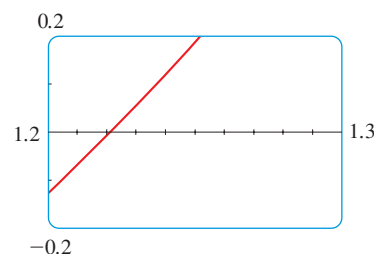


FIGURE 11

We can use a graphing calculator or computer to illustrate the use of the Intermediate Value Theorem in Example 10. Figure 10 shows the graph of f in the viewing rectangle $[-1, 3]$ by $[-3, 3]$ and you can see that the graph crosses the x -axis between 1 and 2. Figure 11 shows the result of zooming in to the viewing rectangle $[1.2, 1.3]$ by $[-0.2, 0.2]$.

In fact, the Intermediate Value Theorem plays a role in the very way these graphing devices work. A computer calculates a finite number of points on the graph and turns on the pixels that contain these calculated points. It assumes that the function is continuous and takes on all the intermediate values between two consecutive points. The computer therefore connects the pixels by turning on the intermediate pixels.

■ Approximating Discontinuous Functions by Continuous Ones

In calculus it's desirable to work with continuous functions, but the functions that arise in biology are often discontinuous. For instance, if $n = P(t)$ is the number of individuals in an animal or plant population at time t , then P is discontinuous whenever a birth or death occurs, as we saw in Figure 4. For a large animal or plant population, however, we can often replace the population function by a continuous function that approximates P . Figure 12 shows the graph of a population function, which is actually a step function, together with the graph of a continuous approximating function.

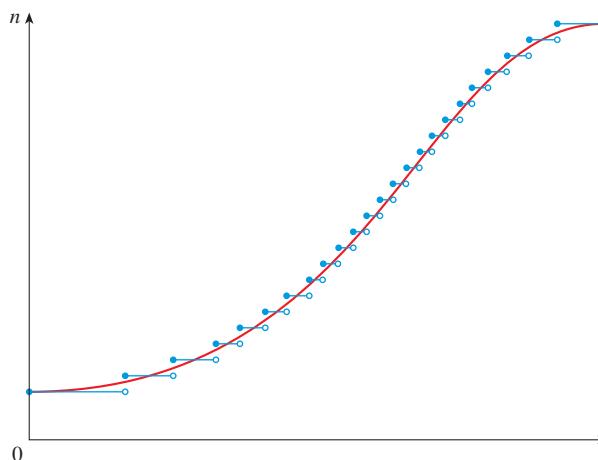


FIGURE 12

With the large numbers involved in the population of the world or HIV replication within a patient, for example, it is reasonable to model the population function with a

continuous function. Indeed, in Section 1.4 we obtained the exponential model

$$P(t) \approx (1436.53) \cdot (1.01395)^t$$

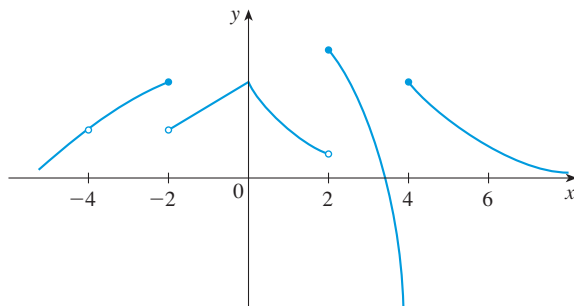
for the world population after 1900 (corresponding to $t = 0$). In general, the exponential functions

$$f(t) = n_0 b^t \quad \text{and} \quad g(t) = n_0 e^{kt}$$

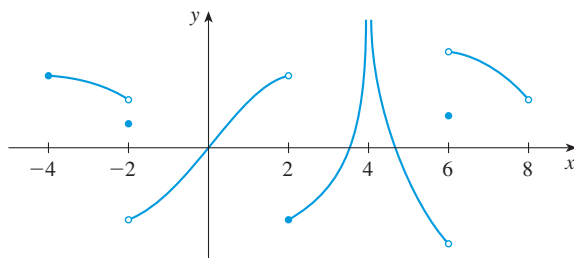
where b and k are constants, are continuous functions that are often used to model population growth, at least in the initial stages.

EXERCISES 2.5

- Write an equation that expresses the fact that a function f is continuous at the number 4.
- If f is continuous on $(-\infty, \infty)$, what can you say about its graph?
- From the graph of f , state the numbers at which f is discontinuous and explain why.
 - For each of the numbers stated in part (a), determine whether f is continuous from the right, or from the left, or neither.



- From the graph of g , state the intervals on which g is continuous.

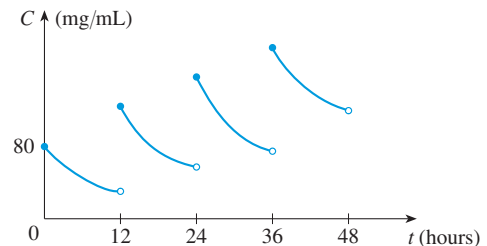


5–8 Sketch the graph of a function f that is continuous except for the stated discontinuity.

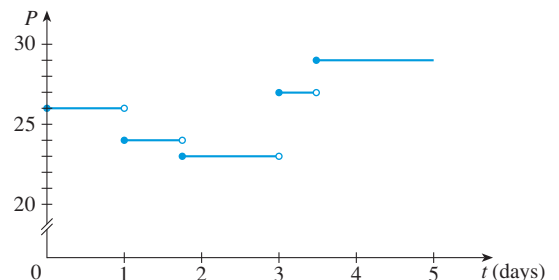
- Discontinuous, but continuous from the right, at 2
- Discontinuities at -1 and 4 , but continuous from the left at -1 and from the right at 4

- Removable discontinuity at 3, jump discontinuity at 5
- Neither left nor right continuous at -2 , continuous only from the left at 2

- Drug concentration** A patient is injected with a drug every 12 hours. The graph shows the concentration $C(t)$ of the drug in the bloodstream after t hours.
 - At what values of t does C have discontinuities?
 - What type of discontinuity does C have?



- Squirrel population** The graph of a population $P(t)$ of squirrels is shown. Identify the discontinuities of P and comment on when and why they occur.



- A parking lot charges \$3 for the first hour (or part of an hour) and \$2 for each succeeding hour (or part), up to a daily maximum of \$10.
 - Sketch a graph of the cost of parking at this lot as a function of the time parked there.
 - Discuss the discontinuities of this function and their significance to someone who parks in the lot.

12. Explain why each function is continuous or discontinuous.
- The temperature in New York City as a function of time
 - The population of New York City as a function of time
 - The temperature at a specific time as a function of the distance due west from New York City
 - The altitude above sea level as a function of the distance due west from New York City
 - The cost of a taxi ride as a function of the distance traveled
13. If f and g are continuous functions with $f(3) = 5$ and $\lim_{x \rightarrow 3} [2f(x) - g(x)] = 4$, find $g(3)$.

14–15 Use the definition of continuity and the properties of limits to show that the function is continuous at the given number a .

14. $f(x) = 3x^4 - 5x + \sqrt[3]{x^2 + 4}$, $a = 2$

15. $f(x) = (x + 2x^3)^4$, $a = -1$

16. Use the definition of continuity and the properties of limits to show that the following function is continuous on the interval $(2, \infty)$.

$$f(x) = \frac{2x + 3}{x - 2}$$

17–20 Explain why the function is discontinuous at the given number a . Sketch the graph of the function.

17. $f(x) = \begin{cases} e^x & \text{if } x < 0 \\ x^2 & \text{if } x \geq 0 \end{cases} \quad a = 0$

18. $f(x) = \begin{cases} \frac{x^2 - x}{x^2 - 1} & \text{if } x \neq 1 \\ 1 & \text{if } x = 1 \end{cases} \quad a = 1$

19. $f(x) = \begin{cases} \cos x & \text{if } x < 0 \\ 0 & \text{if } x = 0 \\ 1 - x^2 & \text{if } x > 0 \end{cases} \quad a = 0$

20. $f(x) = \begin{cases} \frac{2x^2 - 5x - 3}{x - 3} & \text{if } x \neq 3 \\ 6 & \text{if } x = 3 \end{cases} \quad a = 3$

21–26 Explain, using Theorems 4, 5, 6, and 8, why the function is continuous at every number in its domain. State the domain.

21. $R(x) = x^2 + \sqrt{2x - 1}$ 22. $G(x) = \sqrt[3]{x}(1 + x^3)$

23. $L(t) = e^{-5t} \cos 2\pi t$ 24. $h(x) = \frac{\sin x}{x + 1}$

25. $G(t) = \ln(t^4 - 1)$ 26. $F(x) = \sin(\cos(\sin x))$

 **27–28** Locate the discontinuities of the function and illustrate by graphing.

27. $y = \frac{1}{1 + e^{1/x}}$

28. $y = \ln(\tan^2 x)$

29–32 Use continuity to evaluate the limit.

29. $\lim_{x \rightarrow 4} \frac{5 + \sqrt{x}}{\sqrt{5 + x}}$

30. $\lim_{x \rightarrow \pi} \sin(x + \sin x)$

31. $\lim_{x \rightarrow 1} e^{x^2 - x}$

32. **Drug resistance** As we have previously noted (page 25), if p is the current frequency of the resistance gene in a model for the spread of drug resistance, then the frequency in the next generation is

$$\frac{p^2 - 2p}{p^2 - 2}$$

What is the limit of this function as $p \rightarrow \frac{1}{2}$?

33–34 Show that f is continuous on $(-\infty, \infty)$.

33. $f(x) = \begin{cases} x^2 & \text{if } x < 1 \\ \sqrt{x} & \text{if } x \geq 1 \end{cases}$

34. $f(x) = \begin{cases} \sin x & \text{if } x < \pi/4 \\ \cos x & \text{if } x \geq \pi/4 \end{cases}$

35. Find the numbers at which the function

$$f(x) = \begin{cases} x + 2 & \text{if } x < 0 \\ e^x & \text{if } 0 \leq x \leq 1 \\ 2 - x & \text{if } x > 1 \end{cases}$$

is discontinuous. At which of these points is f continuous from the right, from the left, or neither? Sketch the graph of f .

36. The gravitational force exerted by the planet Earth on a unit mass at a distance r from the center of the planet is

$$F(r) = \begin{cases} \frac{GMr}{R^3} & \text{if } r < R \\ \frac{GM}{r^2} & \text{if } r \geq R \end{cases}$$

where M is the mass of Earth, R is its radius, and G is the gravitational constant. Is F a continuous function of r ?

37. For what value of the constant c is the function f continuous on $(-\infty, \infty)$?

$$f(x) = \begin{cases} cx^2 + 2x & \text{if } x < 2 \\ x^3 - cx & \text{if } x \geq 2 \end{cases}$$

38. Suppose that a function f is continuous on $[0, 1]$ except at 0.25 and that $f(0) = 1$ and $f(1) = 3$. Let $N = 2$. Sketch two possible graphs of f , one showing that f might not satisfy the conclusion of the Intermediate Value Theorem and one showing that f might still satisfy the conclusion of the Intermediate Value Theorem (even though it doesn't satisfy the hypothesis).
39. If $f(x) = x^2 + 10 \sin x$, show that there is a number c such that $f(c) = 1000$.
40. Suppose f is continuous on $[1, 5]$ and the only solutions of the equation $f(x) = 6$ are $x = 1$ and $x = 4$. If $f(2) = 8$, explain why $f(3) > 6$.

41–44 Use the Intermediate Value Theorem to show that there is a solution of the given equation in the specified interval.

41. $x^4 + x - 3 = 0$, $(1, 2)$ 42. $\sqrt[3]{x} = 1 - x$, $(0, 1)$
 43. $e^x = 3 - 2x$, $(0, 1)$ 44. $\sin x = x^2 - x$, $(1, 2)$

45–46 (a) Prove that the equation has at least one real root. (b) Use your calculator to find an interval of length 0.01 that contains a root.

45. $\cos x = x^3$ 46. $\ln x = 3 - 2x$

-  **47–48** (a) Prove that the equation has at least one real solution. (b) Use your graphing device to find the solution correct to three decimal places.

47. $100e^{-x/100} = 0.01x^2$ 48. $\sqrt{x-5} = \frac{1}{x+3}$

49. Is there a number that is exactly 1 more than its cube?
50. Show that the function

$$f(x) = \begin{cases} x^4 \sin(1/x) & \text{if } x \neq 0 \\ 0 & \text{if } x = 0 \end{cases}$$

is continuous on $(-\infty, \infty)$.

51. A Tibetan monk leaves the monastery at 7:00 AM and takes his usual path to the top of the mountain, arriving at 7:00 PM. The following morning, he starts at 7:00 AM at the top and takes the same path back, arriving at the monastery at 7:00 PM. Use the Intermediate Value Theorem to show that there is a point on the path that the monk will cross at exactly the same time of day on both days.

Chapter 2 REVIEW

CONCEPT CHECK

- (a) What is a convergent sequence?
(b) What does $\lim_{n \rightarrow \infty} a_n = 3$ mean?
- What is $\lim_{n \rightarrow \infty} r^n$ in the following three cases?
(a) $0 < r < 1$ (b) $r = 1$ (c) $r > 1$
- (a) What is the sum of the finite geometric series $a + ar + ar^2 + \cdots + ar^n$?
(b) If $-1 < r < 1$, what is the sum of the infinite geometric series $a + ar + ar^2 + \cdots + ar^n + \cdots$?
- Explain what each of the following means and illustrate with a sketch.

(a) $\lim_{x \rightarrow a} f(x) = L$	(b) $\lim_{x \rightarrow a^+} f(x) = L$
(c) $\lim_{x \rightarrow a^-} f(x) = L$	(d) $\lim_{x \rightarrow a} f(x) = \infty$
(e) $\lim_{x \rightarrow \infty} f(x) = L$	
- State the following Limit Laws for functions.

(a) Sum Law	(b) Difference Law
(c) Constant Multiple Law	(d) Product Law
(e) Quotient Law	(f) Power Law
(g) Root Law	
- What does the Squeeze Theorem say?
- (a) What does it mean to say that the line $x = a$ is a vertical asymptote of the curve $y = f(x)$? Draw curves to illustrate the various possibilities.
(b) What does it mean to say that the line $y = L$ is a horizontal asymptote of the curve $y = f(x)$? Draw curves to illustrate the various possibilities.
- Which of the following curves have vertical asymptotes? Which have horizontal asymptotes?

(a) $y = x^4$	(b) $y = \sin x$
(c) $y = \tan x$	(d) $y = e^x$
(e) $y = \ln x$	(f) $y = 1/x$
(g) $y = \sqrt{x}$	
- (a) What does it mean for f to be continuous at a ?
(b) What does it mean for f to be continuous on the interval $(-\infty, \infty)$? What can you say about the graph of such a function?
- What does the Intermediate Value Theorem say?

Answers to the Concept Check can be found on the back endpapers.